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Configuring Fibre Channel Protocol on IBM LinuxONE and IBM Z to IBM Storage FlashSystem

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Introduction

IBM® LinuxONE and IBM Z® platforms offer support for open systems Fibre Channel Protocol networks and subsystems, so you can use them for block storage provisioning for Linux and select IBM z/VM® workloads. This IBM Redpaper provides a comprehensive guide to the hardware configuration and enablement of FCP for Linux environments.

This document is intended for IBM Z and LinuxONE system administrators, storage administrators, infrastructure architects, and implementers.

Fibre Channel Protocol overview

Fibre Channel Protocol (FCP) is a modern storage fabric interconnect that provides transport from Server host systems to their Storage devices. IBM FICON® uses the FCP infrastructure and provides a modern replacement for the ESCON interconnect protocol for high-speed mainframe storage connectivity.

The implementation by IBM of FCP on IBM LinuxONE and IBM Z platforms provides access for Linux and z/VM environments to a comprehensive portfolio of storage technologies from diverse vendors. It also capitalizes on the inherent flexibility and robust connectivity options that are characteristic of FCP.

You can use FCP to build a network of FCP switches, hosts, and storage controllers that connect to each other and create a Storage Area Network (SAN). The following definitions are for the ports that are listed in Figure 1 on page 2:

- N_Ports connect host and storage devices to the SAN.
- F_Ports connect the SAN to host and storage devices.
- E_Ports connect SAN switches to other SAN switches.
- L_Ports represent Arbitrated Loop ports and encompass both N_Port and F_Port functions. These port types are relatively uncommon in modern Fibre Channel deployments.

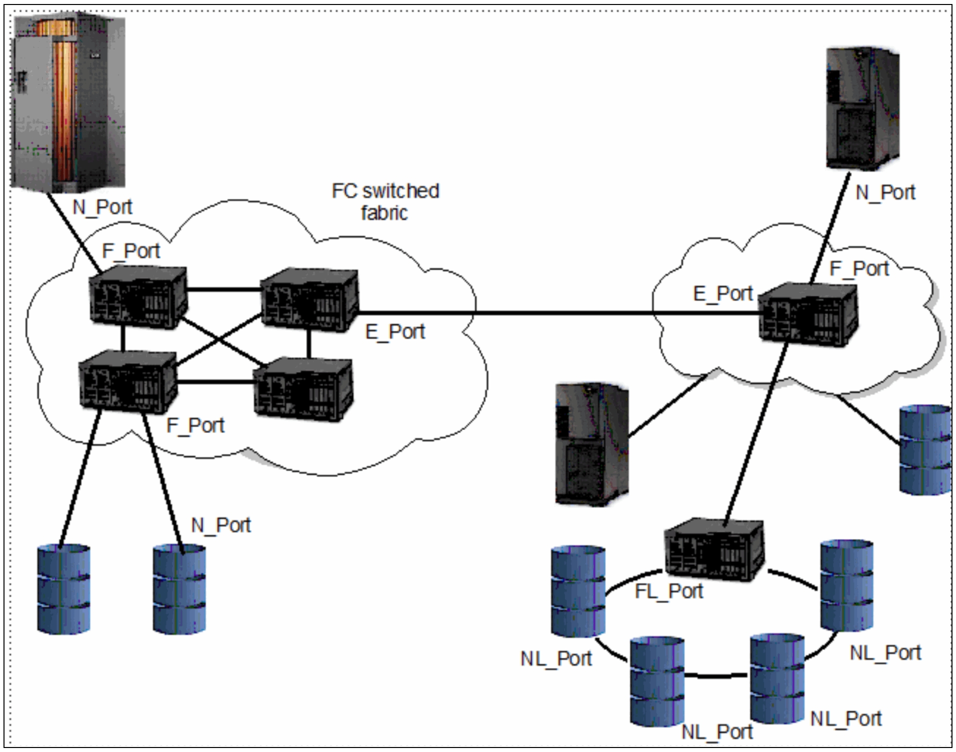


Figure 1 A typical SAN

Devices in the network are identified by World Wide Node Names (WWNN), and individual ports within devices are identified by World Wide Port Names (WWPN). See Figure 2.

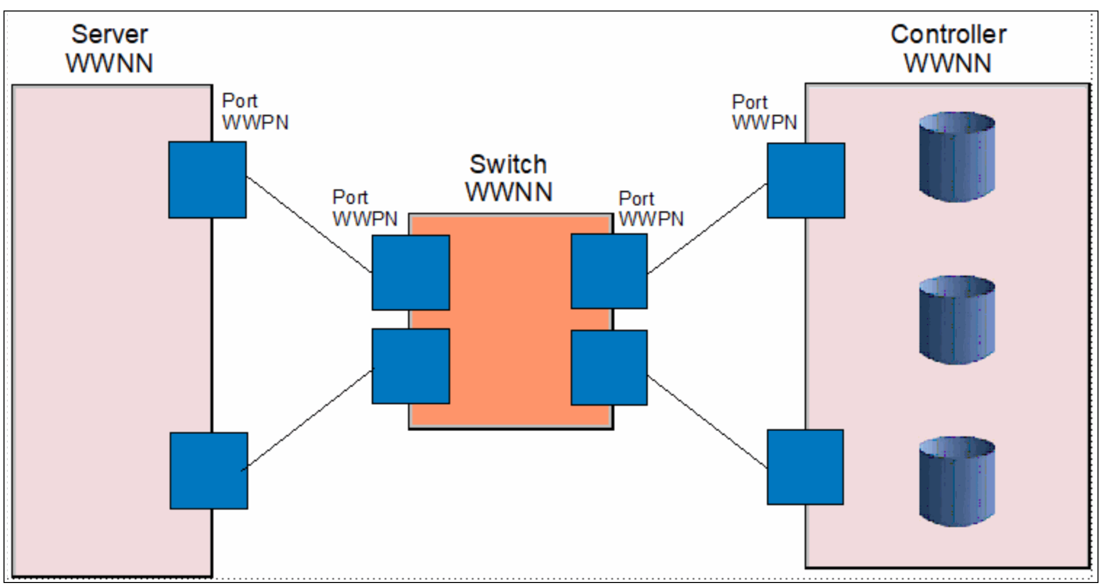


Figure 2 Example of worldwide names

All ports, which include host ports, storage ports, and switch ports, have a globally unique WWPN. To prevent name collisions across device vendors, WWPNs are generated according to rules that are similar to Ethernet MAC addresses.

Access control is managed by lists of WWPNs that are permitted to communicate both in the switch and in the storage controller.

In the switch, the access list is called a zone. A zone contains the list of WWPNs of the ports that can talk to each other.

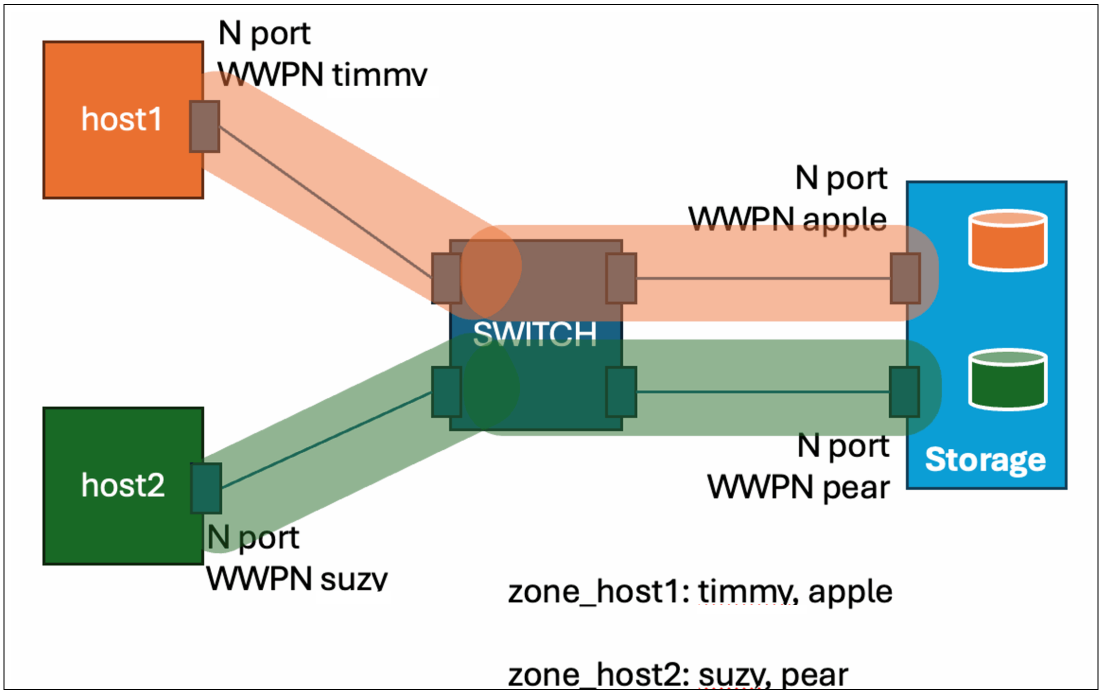


Figure 3 Zones

In Figure 3, there are 2 zones. One zone is named zone_host1, and it permits WWPN timmy WWPN apple to talk to each other. The other zone is named zone_host2, and it permits WWPN suzy and WWPN pear to talk to each other. The switch does not permit any WWPN to converse with or know the existence of a WWPN that it is not in the same zone. The physical ports on the switch where host and storage are connected do not affect zoning. However, these connections significantly impact availability and backplane bandwidth.

Zoning is a fabric level construct and extends across all switches in a multi switch fabric. If SWITCH is a network of several connected switches, the zones are not affected by the internal links. Zones are concerned only with the endpoints.

Storage controllers present their virtual storage volumes to the network as logical units, and are accessed from the host by using their logical unit number (LUN) (Figure 4 on page 4).

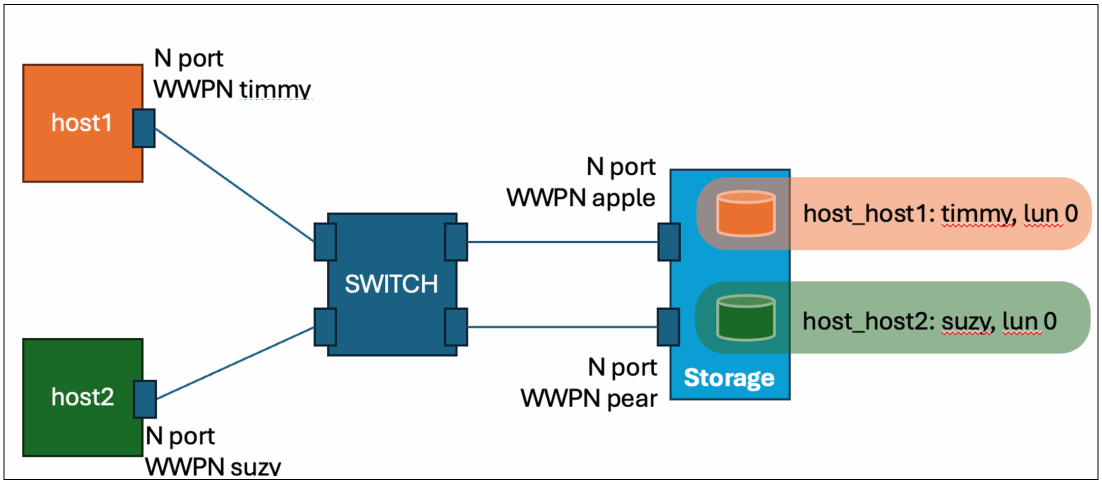


Figure 4 LUNs

In the storage controller, the access list is called a host definition with an associated LUN mapping. It is a list of WWPNs associated with a specific server or virtual machine and the list of virtual volumes in the storage controller that the system is permitted to access. In Figure 4, there is a host definition named host_host1 with the WWPN timmy listed, and a LUN map that associates a volume with that host as LUN 0. There is another host definition named host_host2 with the WWPN suzy listed, and it maps a different volume to that host, but also as LUN 0.

Both hosts have a LUN 0 in the storage controller, but the volumes that are accessed at LUN 0 are different.

A host definition can also be a cluster of systems with multiple WWPNs from all cluster members that you can use to make sure that the LUN numbers are consistent across all the cluster members, if that's important to you.

Some storage controllers, such as the IBM FlashSystem® devices, count the virtual volumes as they map them to a defined host starting at LUN number 0 and counting up.

Other storage controllers, such as the IBM DS8000® class of systems, enumerate the virtual volumes by using a fixed virtual controller and virtual device number that is intrinsic to the virtual volume itself as they map the volumes to a defined host.

Tip: This difference has some implications for configuration. IBM FlashSystem allows administrators to easily assign LUN 0 to specific devices, like boot volumes. However, IBM DS8000 systems do not use LUN 0; they assign identifiers, such as `40104022` (indicating the 22nd virtual volume on logical controller 10).

FCP and FICON comparison

For storage configurations, the comparison of FICON and FCP includes some key differences. In FICON, the host's I/O configuration describes the entire storage network, from the host through the switches to the storage systems. The network topology is defined within this I/O configuration. Defined control units (CUs) and devices directly correspond to logical control units and devices on the storage controller.

For example, a CU 7000 definition on channel path IDs (CHPIDs) 40 and 41 that have 16 devices with IDs of 7000–700F maps directly to a specific logical control unit on an extended count-key-data (ECKD) storage server with 16 corresponding devices. The device with address 7000 typically represents the same 3390 storage volume across all LPARs that can access the volume.

Also, the path from the host port to the storage device is defined in the I/O configuration by using the physical port numbers of the switch. The FICON I/O configuration is like a roadmap that describes how to reach the storage:

1. Take CHPID 20 to Switch 33.
2. Leave Switch 33 through port 09 to get to the DS8000.
3. In the DS8000, take the 9th right from the roundabout to get to LCU 9 in the DS8000.

This means that if you move a link between the storage controller and the switch to a different port on the switch, the defined link from the Z system to the storage controller goes offline because the exit port from the switch is now unplugged.

FCP functions differently. FCP control units and devices on the host behave more like open systems adapter (OSA) devices than physical disks. From the host's perspective, the FCP device is an interface that is used by the zfcp driver for network communication. The SAN switches and storage controllers manage all network routing and storage interactions. Therefore, even if FCP devices in different LPARs (for example, device 7000 in LPAR 1 and device 7000 in LPAR 2) share a physical port in the I/O configuration, they are treated as distinct entities.

N_Port ID virtualization

In a SAN fabric, N_Ports designate the host or storage endpoints. If a single physical port is used by multiple logical entities with different needs, the SAN requires a mechanism to distinguish between these entities to ensure proper zoning and LUN mapping.

N_Port ID virtualization (NPIV) implements a virtualization technique that generates distinct virtual World Wide Port Names (WWPNs) for each logical entity that uses a shared physical SAN port. This virtualization is essential for ensuring the proper operation of zoning and storage masking mechanisms.

In Figure 5 on page 6, host1 and host2 are LPARs on a LinuxONE machine that share access to an FCP Channel without NPIV active. Without NPIV, there is just the physical WWPN of the port, which host1 and host2 are sharing. The switch cannot tell the difference between traffic from host1 and host2 because it all comes from WWPN suzy, so all users of that port can access everything that WWPN suzy is zoned to.

The same problem happens in the Storage controller. The Storage controller sees only I/Os coming from WWPN suzy, so they can access the resources that are mapped to suzy.

This is not unique to IBM Z® and LinuxONE systems, this problem applies to basically any hypervisor platform.

After you activate NPIV on FCP ports, you can generate unique WWPNs for each LPAR's devices that share the physical port.

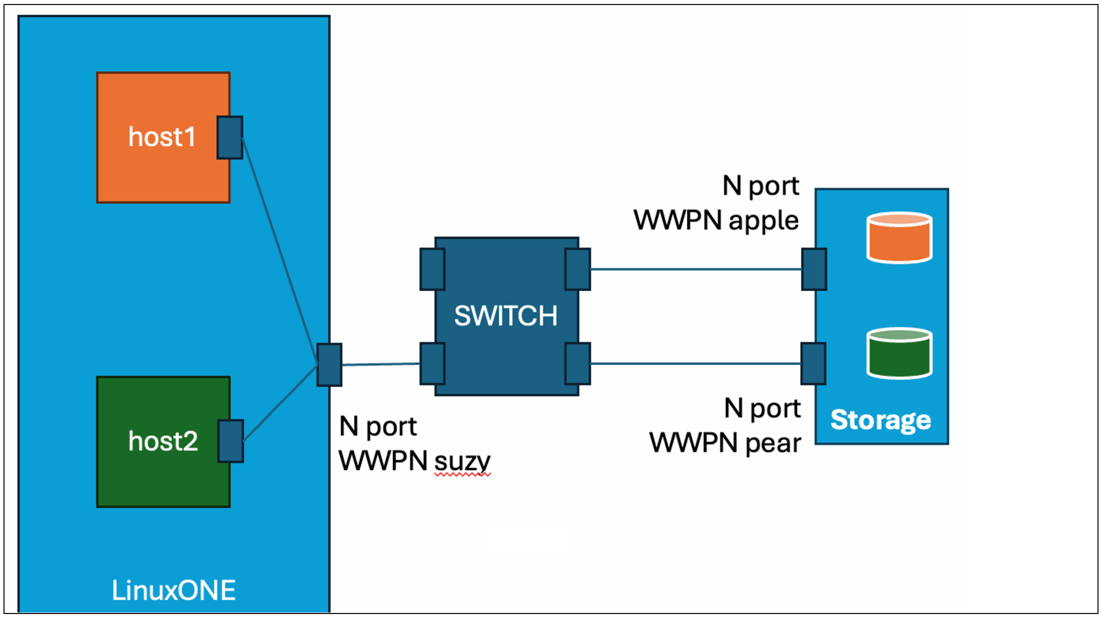


Figure 5 LPARs on a LinuxONE machine, sharing access to an FCP Channel without NPIV active

Figure 6 shows an activated NPIV on that port for the host1 and host2 LPARs. Now, I/O traffic coming from host1 is seen by the SAN as coming from WWPN carrot, and I/O traffic coming from host2 is seen as coming from WWPN potato. The physical port WWPN is still suzy, but it is no longer used for zoning or LUN mapping.

When NPIV is used, the SAN fabric recognizes the FCP logical devices presented to LPAR host1 and LPAR host2 as separate entities. Therefore, this requires distinct zoning in the SAN switches and an individual host definition for each LPAR on the storage controller to provide access to their specific LUNs.

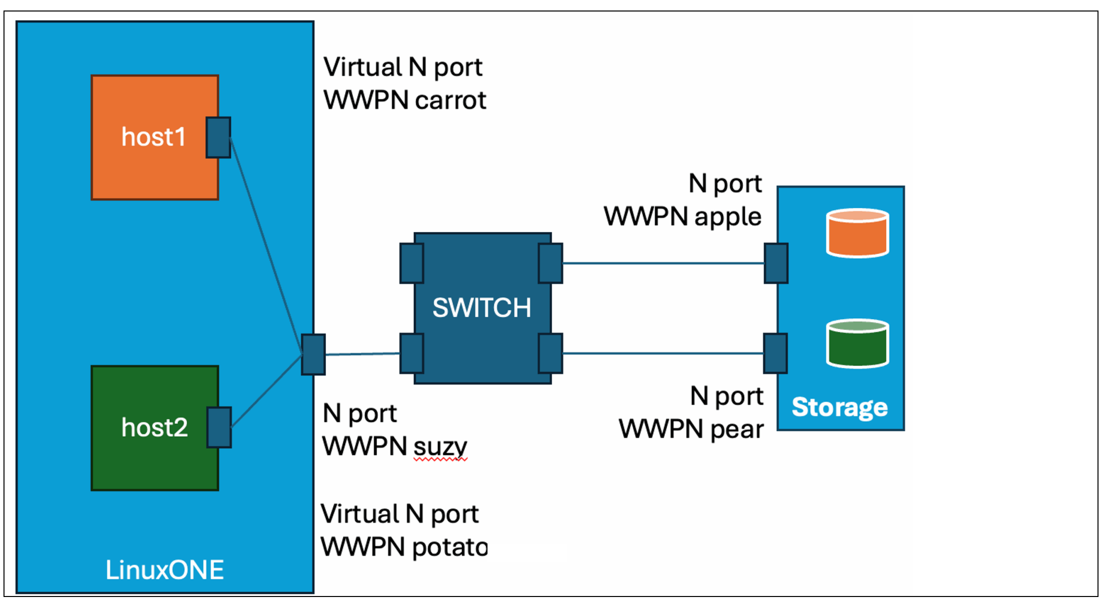


Figure 6 NPIV is activated for the host1 and host2 LPARs

Supported storage services

The IBM Storage dev/test organization is responsible for testing and determining what IBM Storage products are supported for attachment to LinuxONE and IBM z Systems®.

The IBM System Storage Interoperation Center (SSIC) that is provided by IBM Storage helps determine which storage devices are supported for connection to different server platforms, such as IBM LinuxONE and Z. The SSIC also provides information on compatible switch interconnections and tested firmware bundles.

Storage vendors are responsible for testing and confirming compatibility between their products and IBM LinuxONE or IBM Z. To determine whether a specific storage product is supported, consult the storage vendor.

Defining FCP devices on IBM Z and LinuxONE

Example 1 shows a simple example of defining FCP devices in IOCP that are shared across all LPARs on a LinuxONE Emperor system with 6 Logical Channel Subsystems running in classic (not DPM) mode:

- PCHID 180 is shared by all LPARs as CHPID 40.
- CHPID 40 connects to CU A002, which provides devices A000-A002.
- PCHID 181 is shared by all LPARs as CHPID 45.
- CHPID 45 connects to CU A302, which provides devices A300-A302.
- PCHID 100 is shared by all LPARs as CHPID 46.
- CHPID 46 connects to CU A102, which provides devices A100-A102.
- PCHID 101 is shared by all LPARs as CHPID 4C.
- CHPID 4C connects to CU A202, which provides devices A200-A202.

Example 1 Defining FCP devices in IOCP

```
CHPID PATH=(CSS(0,1,2,3,4,5),40),SHARED,PCHID=180,TYPE=FCP
CHPID PATH=(CSS(0,1,2,3,4,5),45),SHARED,PCHID=181,TYPE=FCP
CHPID PATH=(CSS(0,1,2,3,4,5),46),SHARED,PCHID=100,TYPE=FCP
CHPID PATH=(CSS(0,1,2,3,4,5),4C),SHARED,PCHID=101,TYPE=FCP

CNTLUNIT CUNUMBR=A002,                                         *
      PATH=((CSS(0),40),(CSS(1),40),(CSS(2),40),(CSS(3),40),(C*
      SS(4),40),(CSS(5),40)),UNIT=FCP
IODEVICE ADDRESS=(A000,003),CUNUMBR=(A002),UNIT=FCP

CNTLUNIT CUNUMBR=A102,                                         *
      PATH=((CSS(0),46),(CSS(1),46),(CSS(2),46),(CSS(3),46),(C*
      SS(4),46),(CSS(5),46)),UNIT=FCP
IODEVICE ADDRESS=(A100,003),CUNUMBR=(A102),UNIT=FCP

CNTLUNIT CUNUMBR=A202,                                         *
      PATH=((CSS(0),4C),(CSS(1),4C),(CSS(2),4C),(CSS(3),4C),(C*
      SS(4),4C),(CSS(5),4C)),UNIT=FCP
IODEVICE ADDRESS=(A200,003),CUNUMBR=(A202),UNIT=FCP

CNTLUNIT CUNUMBR=A302,                                         *
      PATH=((CSS(0),45),(CSS(1),45),(CSS(2),45),(CSS(3),45),(C*
      SS(4),45),(CSS(5),45)),UNIT=FCP
```

```
IODEVICE ADDRESS=(A300,003),CUNUMBR=(A302),UNIT=FCP
```

For example, although device A000 is visible as an FCP device on all LPARs, A000 on LPAR LP01 is distinct from A000 on LPAR LP17. FCP devices are not DASD volumes. They provide only a network interface for SAN communication.

Consult Appendix F "*Machine Limits and Rules*" in Input/Output Configuration Program User's Guide for ICP IOCP and a table that is named Constant Machine Limits, which at the time of writing is Table 16. The table includes a value for `Maximum devices or subchannels per FCP channel path`, which for the test system is `480`. That means you can configure a maximum of 480 FCP devices on a single FCP Channel.

An example configuration

Consider the I/O configuration and the "Maximum devices or subchannels per FCP channel path" limit of 480 for channel 180. A common misconception is that three devices are configured. However, the actual number depends on the number of LPARs that are using the channel. Given a maximum of 85 LPARs for this machine class, the upper limit is 255 (3 devices/LPAR * 85 LPARs).

However, a note in the table states, "If an FCP channel path is dedicated to a logical partition, the maximum number of devices is 254." Because this channel is not dedicated, then this limitation does not apply.

Machine-specific limits, detailed in the Machine Limits (by Machine Type) table, must also be considered (Figure 7).

Machine Limit	Machine type (Note "1" on page 315)											
	2817,1	2818,1	2827,1	2828,1	2964,1	2965,1	3906,1	3907,1	8561,1	8562,1	3931,1	3932,1
Maximum FCP channel paths	256	160	256	160	256	160	256	128	384	128	384	96

Figure 7 Machine limits (by Machine type)

As shown in Figure 7, the maximum number of configurable FCP ports varies by machine type. The example configuration uses only four ports, which is within the limits for the machine type.

In the document Input/Output Configuration Program User's Guide for ICP IOCP, locate the table shown in Figure 8 on page 9, which at the time of writing is Table 21.

Table 21. FCP Channel Path Limits		
Limit or Recommendation	**FICON Express 8/8S**	**FICON Express 16S/16S+/ 16SA/32SA/32S**
Maximum configurable devices or subchannels per FCP channel path	480	480
Recommended maximum number of active NPIV-enabled subchannels per FCP channel path	32	64 or 128 (Note "11" on page 316)
Maximum number of remote N_Ports per FCP channel path	512	1024
Maximum concurrent I/O operations per FCP channel path	960	1528
Maximum number of LUNs that can be open at the same time	4096	8192

Figure 8 FCP Channel Path Limits table

Note 11 states, "A maximum of 128 NPIV-enabled subchannels are supported on 3932 and 3931 models (with an LMC patch installed) for FICON Express32S. Otherwise, a maximum of 64 NPIV-enabled subchannels are supported." LMC is the abbreviation for Licensed Machine Code.

Therefore, whether the example configuration exceeds the supported limits depends on the number of LPARs sharing the devices and the specific technologies in use on both the processor and switch sides.

Regarding the processor side, the machine types are z16/LinuxONE v.4, and the adapters are those shipped with that system. Assuming a z16 configuration with 16S+ cards and device sharing across 85 LPARs, the limit is exceeded. However, with only 21 LPARs, the number of devices is calculated to be 63 devices:

```
21 LPARs * 3 devices/LPAR = 63 devices
```

Therefore, the configuration remains within the most restrictive NPIV limit for 16S+ cards.

Regarding switch support for host adapter NPIV, each vendor publishes documentation detailing the specific capabilities and limits of their devices. Some models support a maximum of 64 NPIV addresses per port, and others support up to 256. Some models might not support NPIV-enabled host ports at all. Consult the relevant switch documentation to determine the NPIV capabilities and limits of your specific switches. If a switch supports only 32 NPIV addresses per port, this becomes the effective limit for that link.

It is important to address the ambiguity surrounding the Recommended Maximum of 64 or 128 NPIV-enabled devices. IBM product testing generally shows that 64 NPIV-enabled devices per channel are compatible with most director-class switch chassis that explicitly support NPIV host ports.

Although configuring more than 64 NPIV devices per FCP channel is technically possible, its suitability depends on the specific workload characteristics. A definitive hard limit can be a challenge because of the recovery processes triggered by channel failures, such as cable or optic failures. During channel recovery, each NPIV address undergoes a re-login process to both the switch and the storage. Although 64 concurrent recoveries are typically handled effectively by supported switches, 128 concurrent recoveries, although achievable, require meticulous planning of target port distribution within zones and careful consideration of the "FCP Channel Path Limits" table. Configurations exceeding 128 concurrent recoveries, even with switches that support 256 NPIV addresses per port, can present operational challenges. These challenges can manifest as incomplete fabric logins for some NPIV links, switch blade firmware crashes, or other unpredictable behavior detrimental to enterprise storage network stability.

Returning to the example configuration, if it is reduced to one FCP device per channel, it might be shared across all 85 potential LPARs. With additional limits on sharing, you can share channels across 64 LPARs if the switch ports support 64 NPIV addresses per port.

If a configuration that exceeds 64 NPIV addresses per port is considered, rigorous testing is mandatory. This testing must replicate the scale and concurrent IOPS of the production environment and include cable pull tests performed under peak workload conditions to validate the clean recovery of the switch and FCP channel NPIV configuration. Because of the numerous variables involved, IBM cannot provide a specific support statement for configurations exceeding 64 NPIV addresses per port beyond recommending thorough and comprehensive testing, including diverse failure recovery scenarios under at least 200% load, before production deployment.

When you design the SAN configuration, consider all relevant FCP channel limits. Exceeding these limits can result in unpredictable connectivity issues that can be difficult to diagnose.

In this discussion, assume that all concerns are addressed and that the FCP connectivity is designed to provide each LPAR with the required number of FCP devices (whether zero or many) and adheres to all applicable limits.

Enabling NPIV on FCP channels

When a valid I/O configuration is active on the system, enable NPIV on the FCP channels. Unless there are specific, atypical requirements, enabling NPIV on all FCP channels is recommended. For example, in IBM internal comprehensive test configurations that encompass multiple CPCs, switches, and storage subsystems, NPIV is enabled on all but one FCP channel, which is intentionally configured without NPIV for demonstration purposes.

If NPIV is not enabled for an LPAR's devices, the following procedure activates it, and includes the following steps:

1. Configure the channel offline from the specific LPAR in which you want NPIV enabled.
2. Configure NPIV active on that channel for that LPAR.
3. Configure the channel online to the LPAR in which you want NPIV enabled.

Configuring the FCP channel offline from one specific LPAR

Perform the following steps:

1. From the top Hardware Management Console (HMC) menu select your Central Processor Complex (CPC).

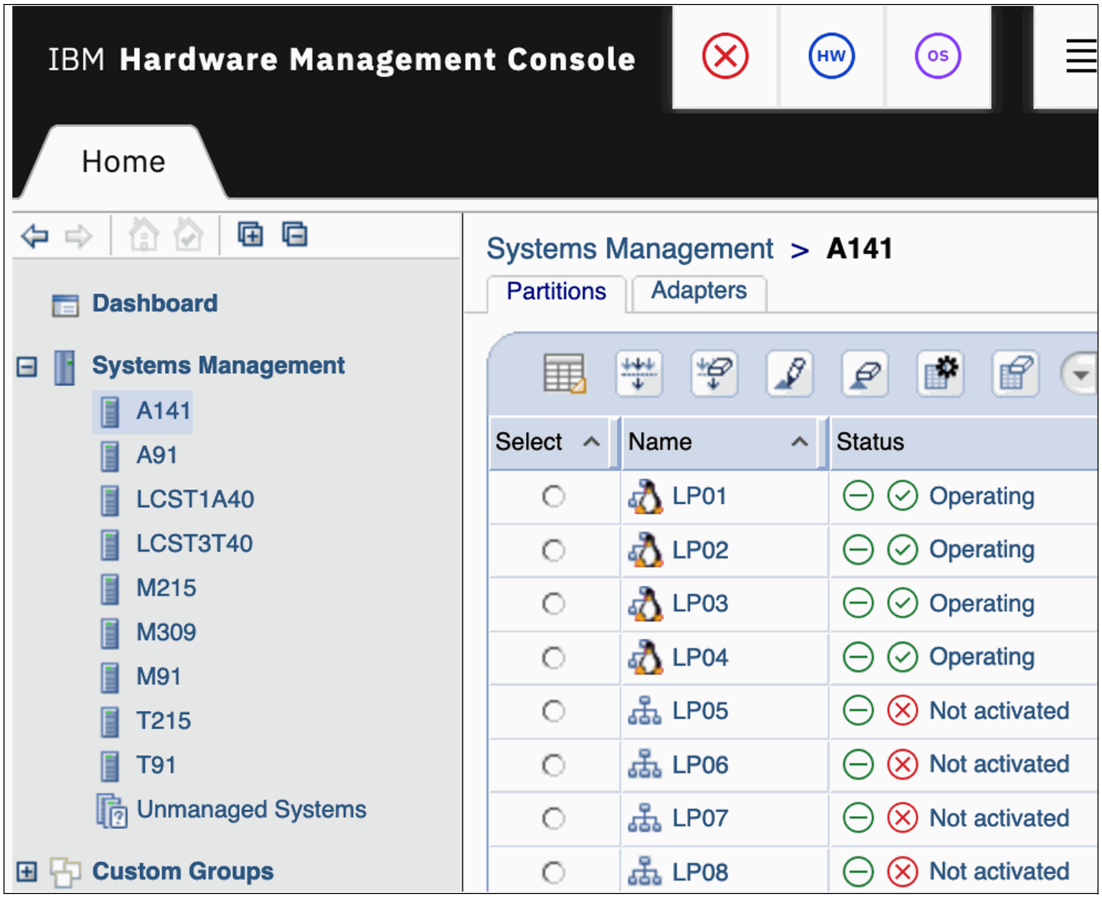


Figure 9 Select your CPC

2. Open the **Recovery** menu on the bottom center pane and select **Single Object Operations** to open the Service Element interface for the selected CPC (Figure 10).

Figure 10 Single Object Operations

> **Note:** The use of Single Object Operations is required only for z14 / LinuxONE II systems. For IBM z15® and IBM z16™ systems, you can select the **Adapters** tab on the HMC view in Figure 9 with the CPC selected and jump to the **Select the Channel ID** step (step 3).

The Service Element window opens in a new browser window (Figure 11 on page 12).

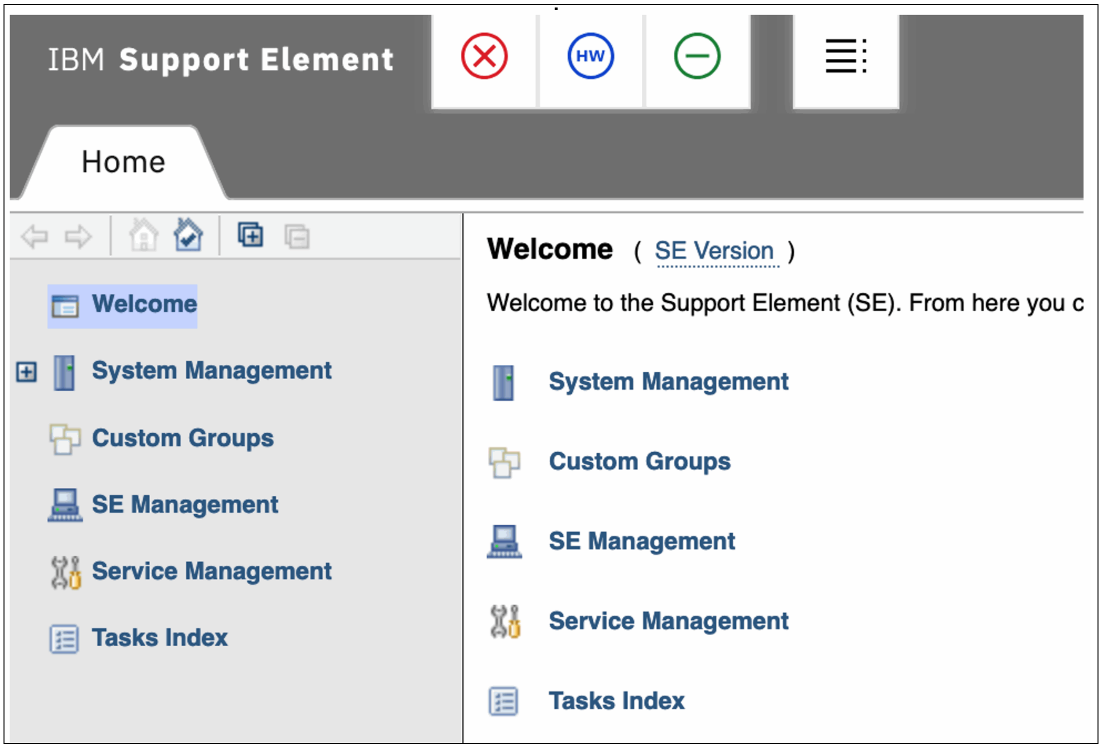


Figure 11 Service Element window

3. In the System Management menu, navigate to the **CPC** menu and select the **Channels** view. Locate the channel ID for the FCP channel requiring NPIV activation for a specific LPAR and select it. This view is similar to the Adapters tab on the HMC for z15 and z16 systems (Figure 12).

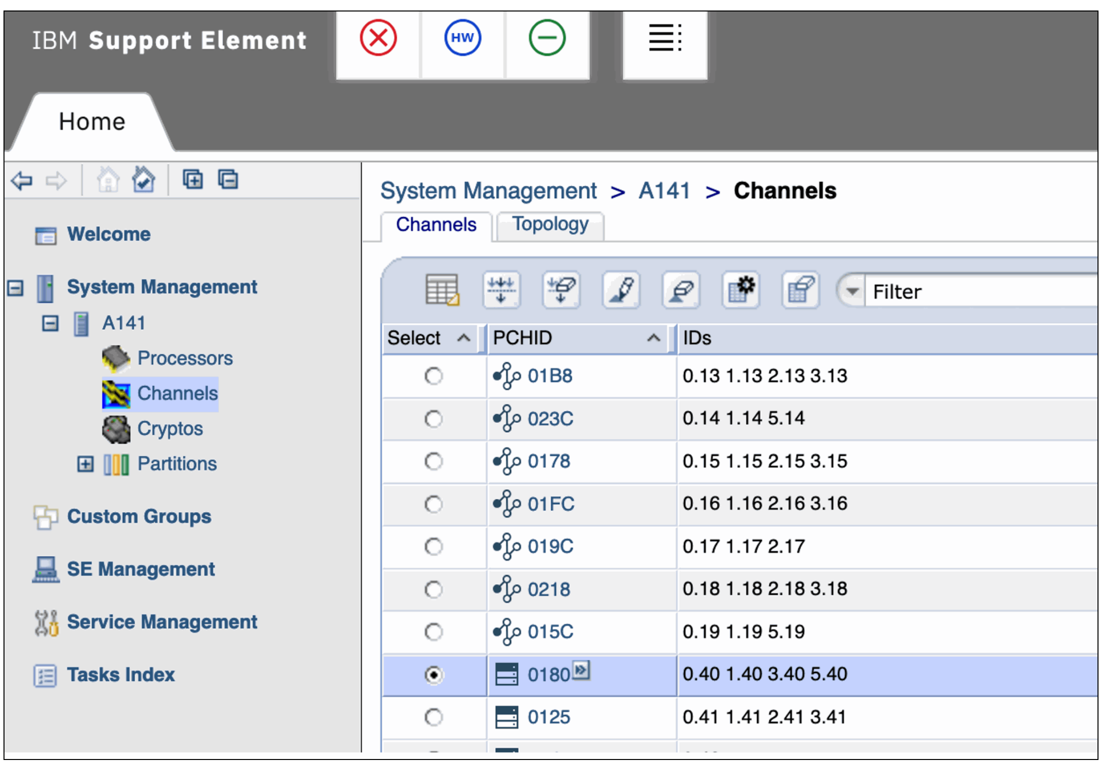


Figure 12 Channels view

4. With the Channel selected, open the **CHPID Operations** menu in the bottom center pane and select **Configure On/Off** (Figure 13).

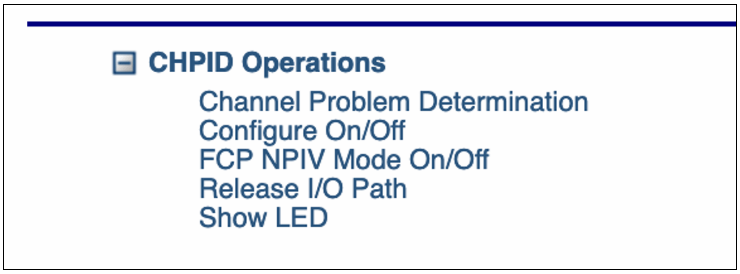


Figure 13   Select Configure On/Off

5. A listing of every LPAR that shares this channel opens. Scroll to and select the entry for the LPAR for which you want to set the channel Offline (Figure 14).

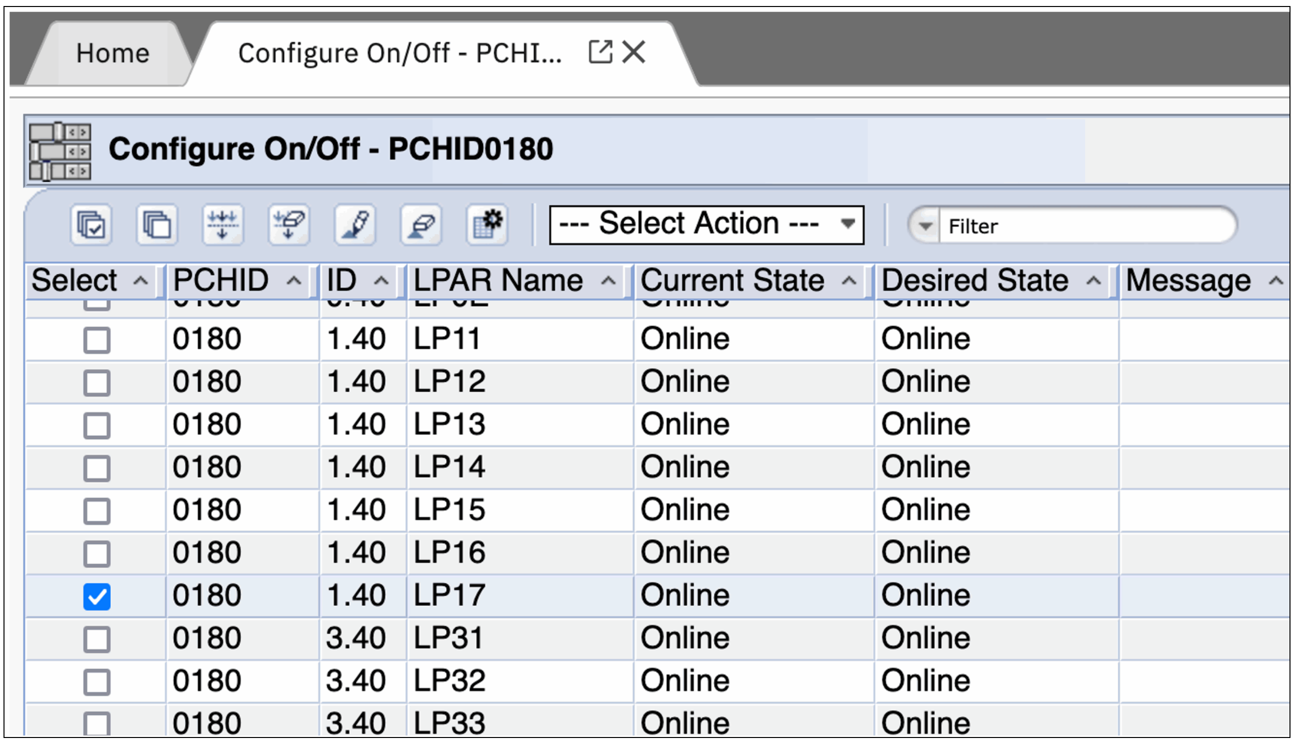


Figure 14   select the entry for the LPAR you want to configure the channel Offline from

6. In the **Select Action** drop-down menu, select the **Toggle** option. Avoid the **Toggle All…** option because it deactivates the channel for all LPARs, which is disruptive if any LPARs are actively using it. Toggle only the CHPID associated with the selected LPAR unless it is confirmed that no LPARs are using the channel (Figure 15 on page 14).

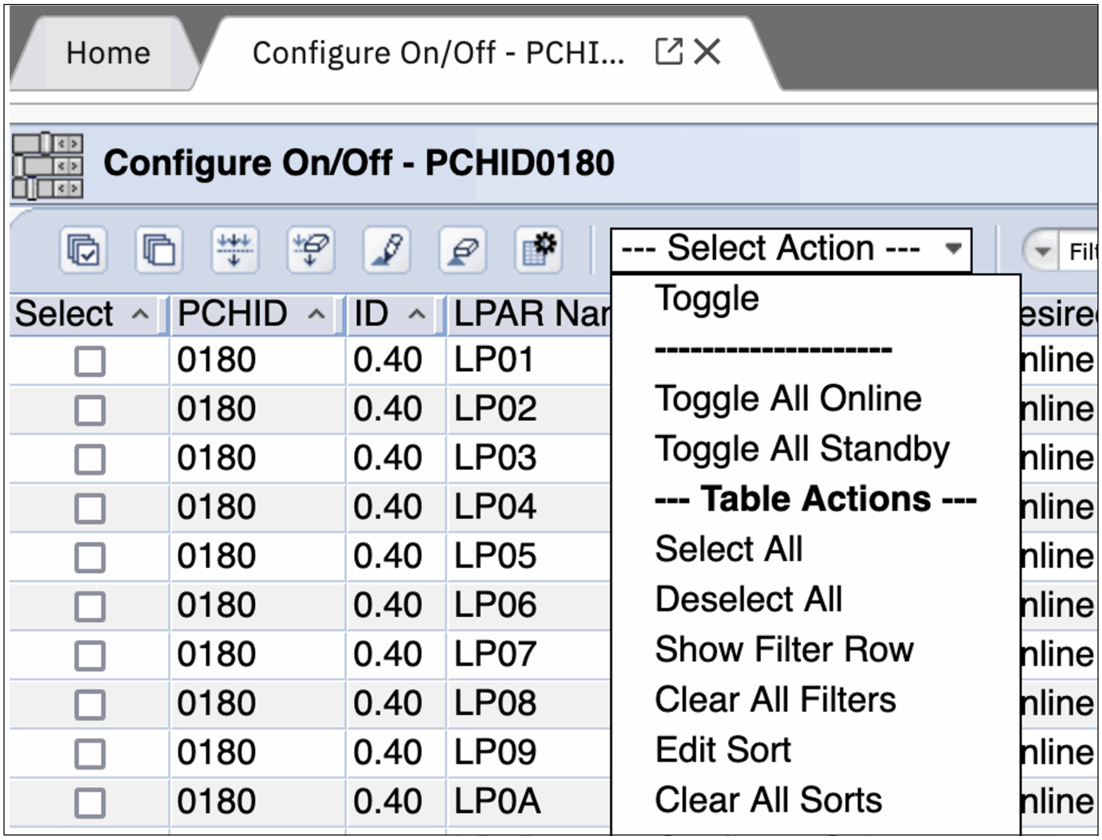


Figure 15 Select the Toggle option

7. Verify that the wanted LPAR CHPID is the only one selected with a changed Desired State. Click **OK** at the bottom of the panel (Figure 16).

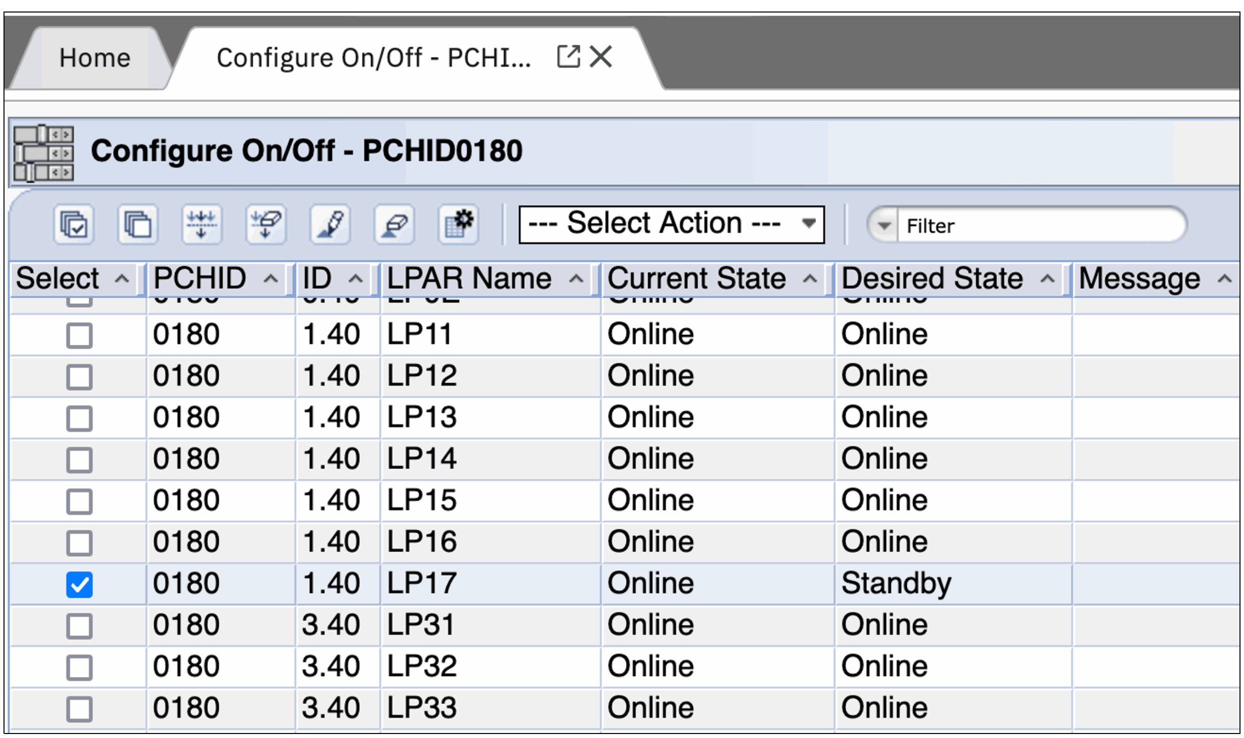


Figure 16 Verify that the wanted LPAR CHPID is the only one selected with a changed Desired State

8. When the confirmation pane opens, verify that you are configuring only the intended LPARs CHPID and click **Yes** (Figure 17).

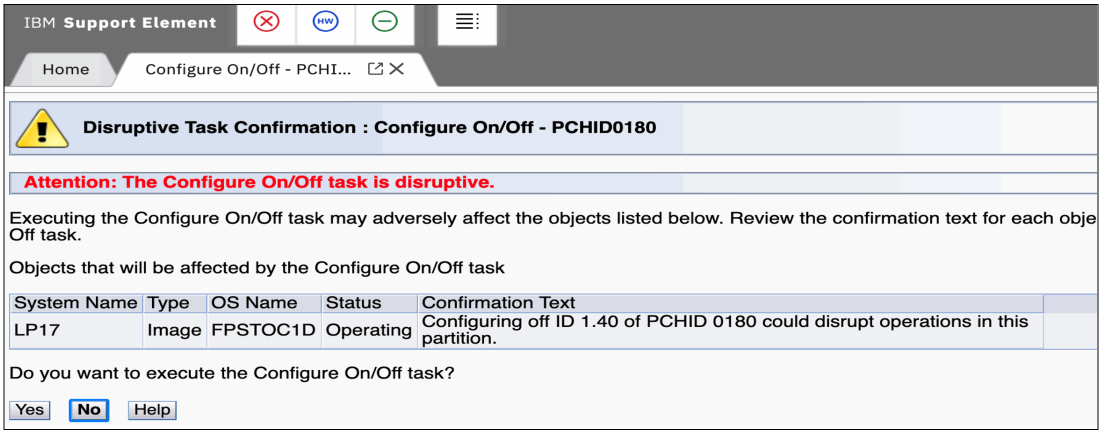


Figure 17 Confirmation pane

Configuring NPIV active on that channel for that LPAR

The CHPID is now Offline for the LPAR. You can enable NPIV by performing the following steps:

1. From the channels view, make sure that the correct Channel is still selected.

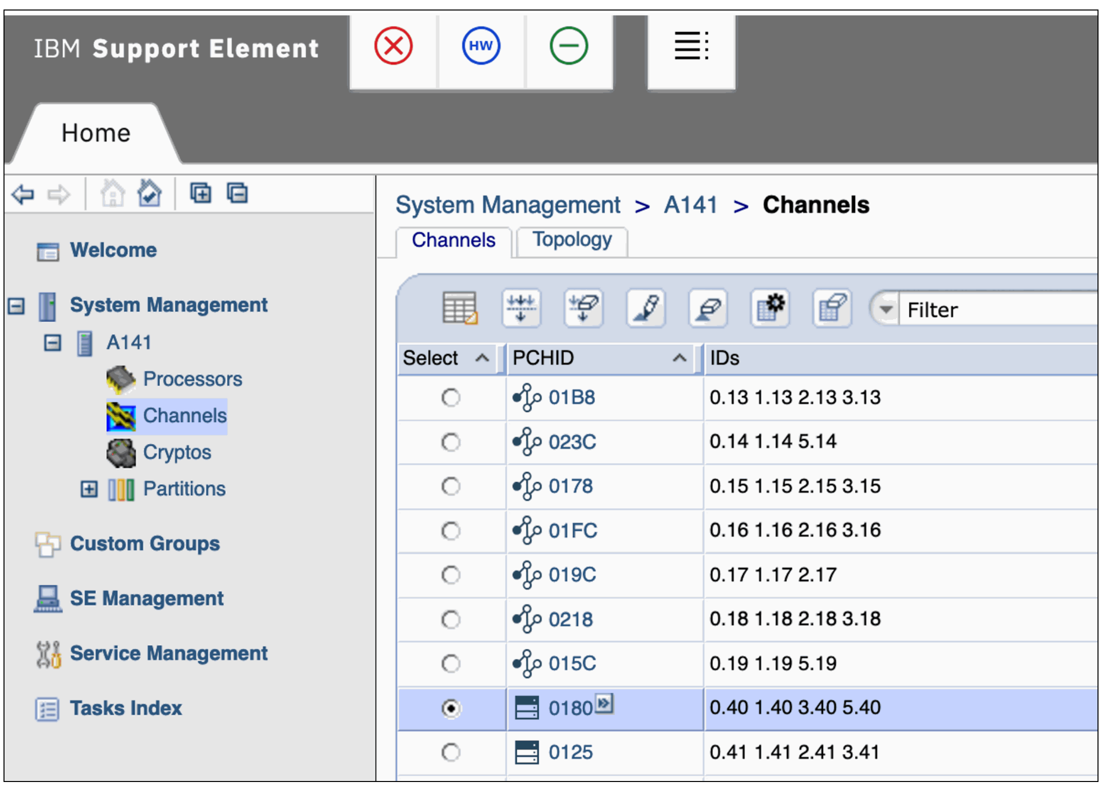


Figure 18 IBM Support Element menu

2. Select the **FCP NPIV Mode On/Off** entry from the CHPID Operations menu in the bottom center pane (Figure 19).

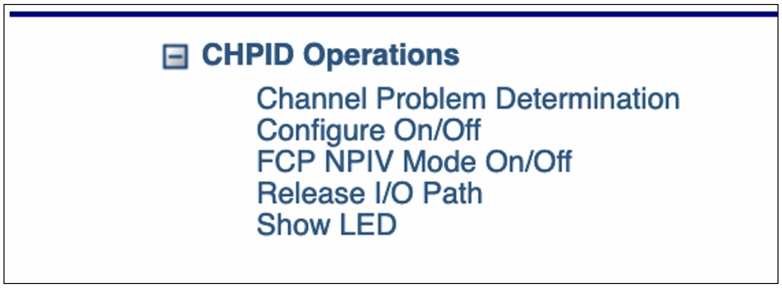


Figure 19 Select the FCP NPIV Mode On/Off entry

3. When the FCP NPIV Mode On/Off tab opens, scroll to and select the **NPIV Mode Enabled** checkbox on the LPAR's row for this CHPID (Figure 20).

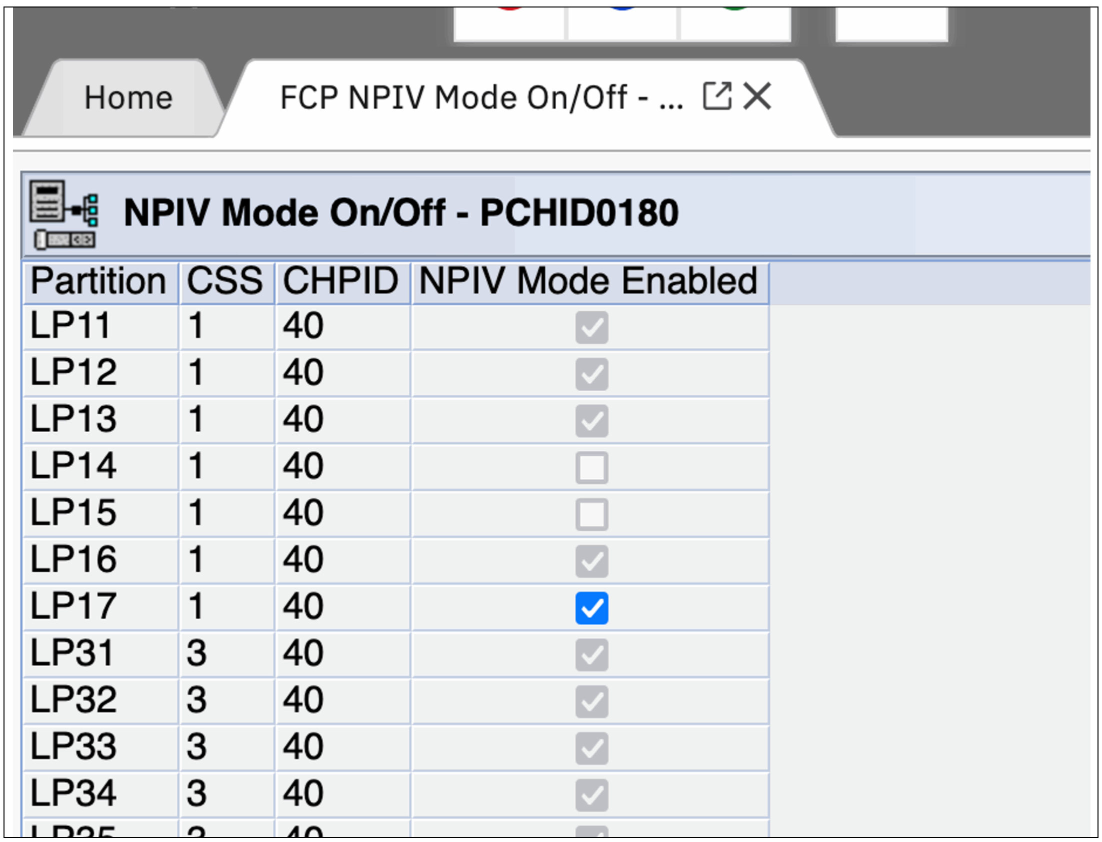


Figure 20 Select the NPIV Mode Enabled checkbox

4. If you do not first set LP17's CHPID 40 to offline, then the checkbox is grayed out, and you cannot select it. Click the **Apply** button at the bottom of the pane to make the change, then click the **Cancel** button to close out of the panel.

Configure the channel online to the LPAR

NPIV is active on the CHPID for the LPAR. Set the status of the CHPID to online by performing the following steps:

1. Open the Channels listing (Figure 21).

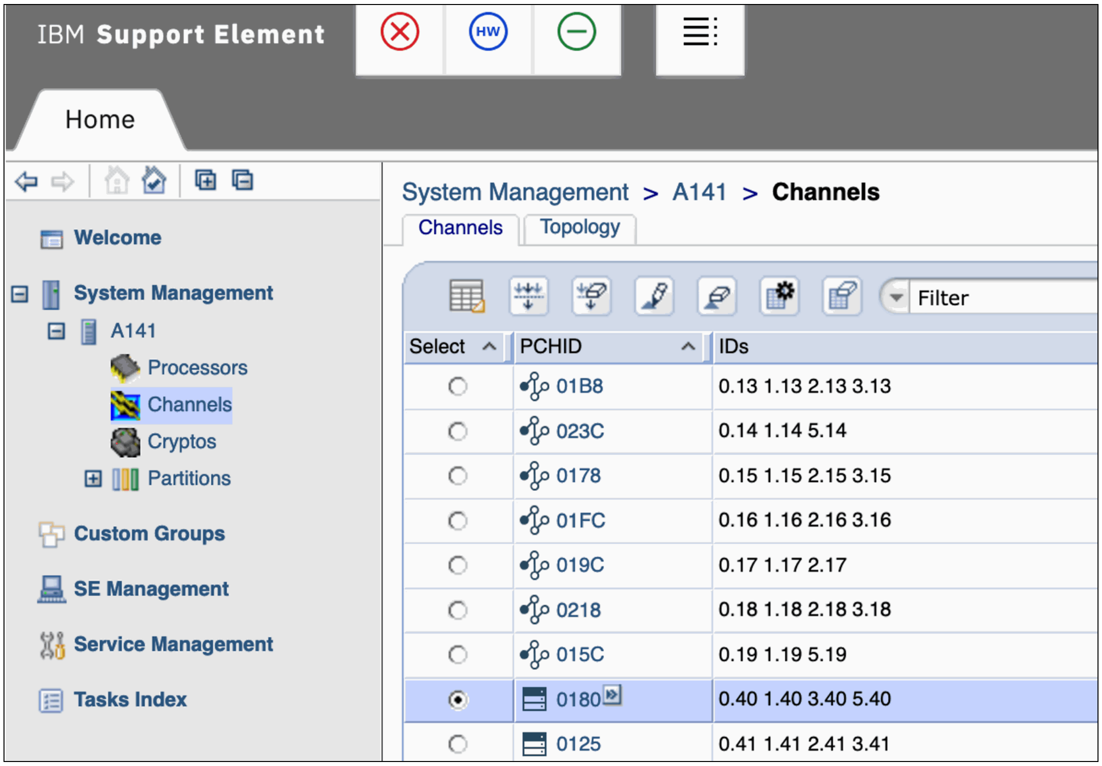


Figure 21 IBM Support Element view

2. Select the **CHPID Operations** menu in the bottom center pane and select **Configure On/Off** (Figure 22).

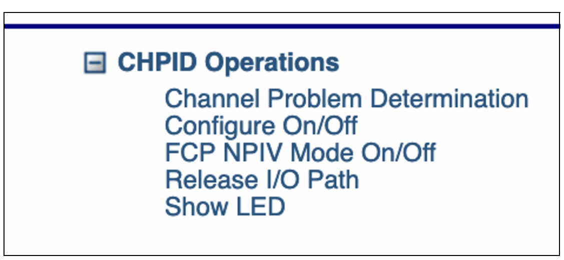


Figure 22 Select Configure On/Off

3. Locate and select the entry corresponding to the LPAR you want to bring back online (Figure 23 on page 18).

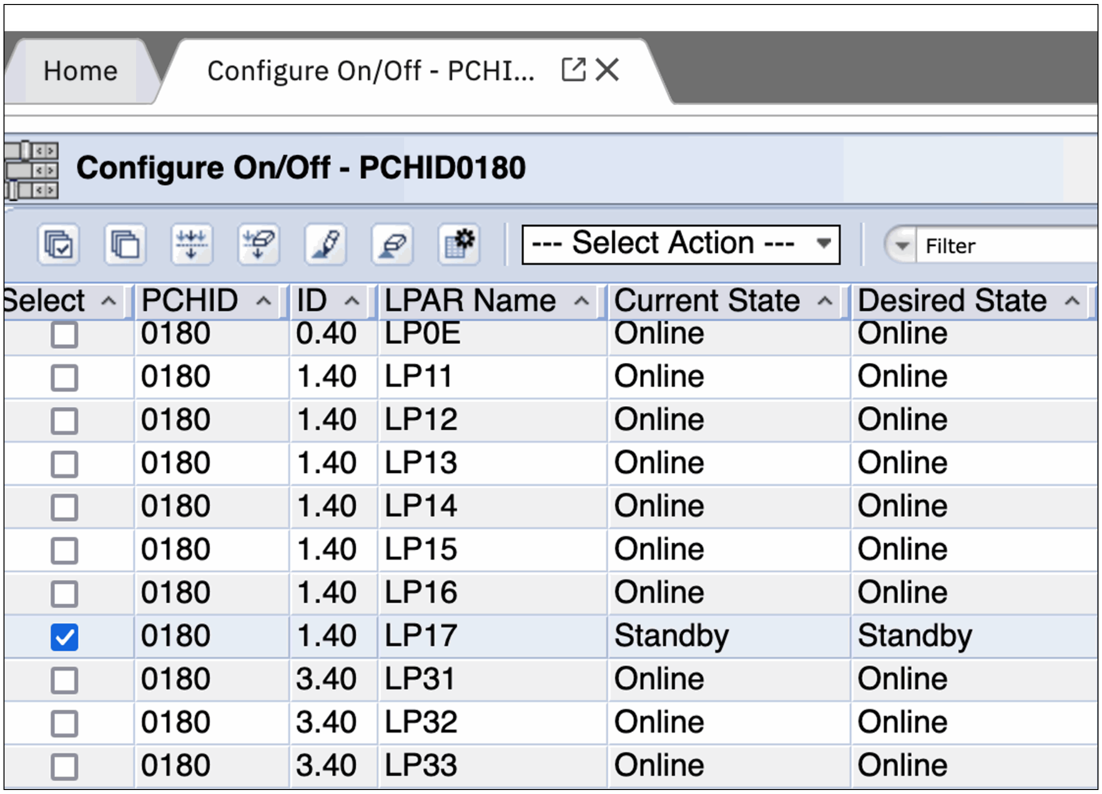


Figure 23 Locate and select the entry corresponding to the LPAR you want to bring back online

4. Select **Toggle** from the drop-down menu. Toggle only the one CHPID for the selected LPAR (Figure 24).

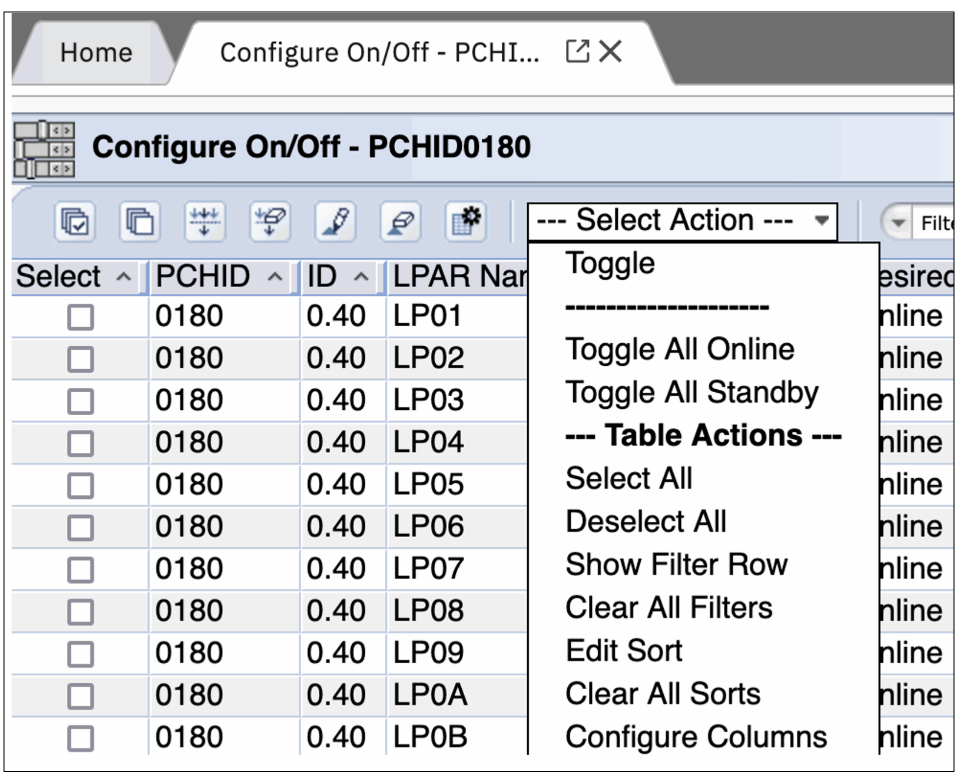


Figure 24 Select Toggle from the drop-down menu

5. The panel refreshes to show the wanted change (Figure 25). Click **OK**.

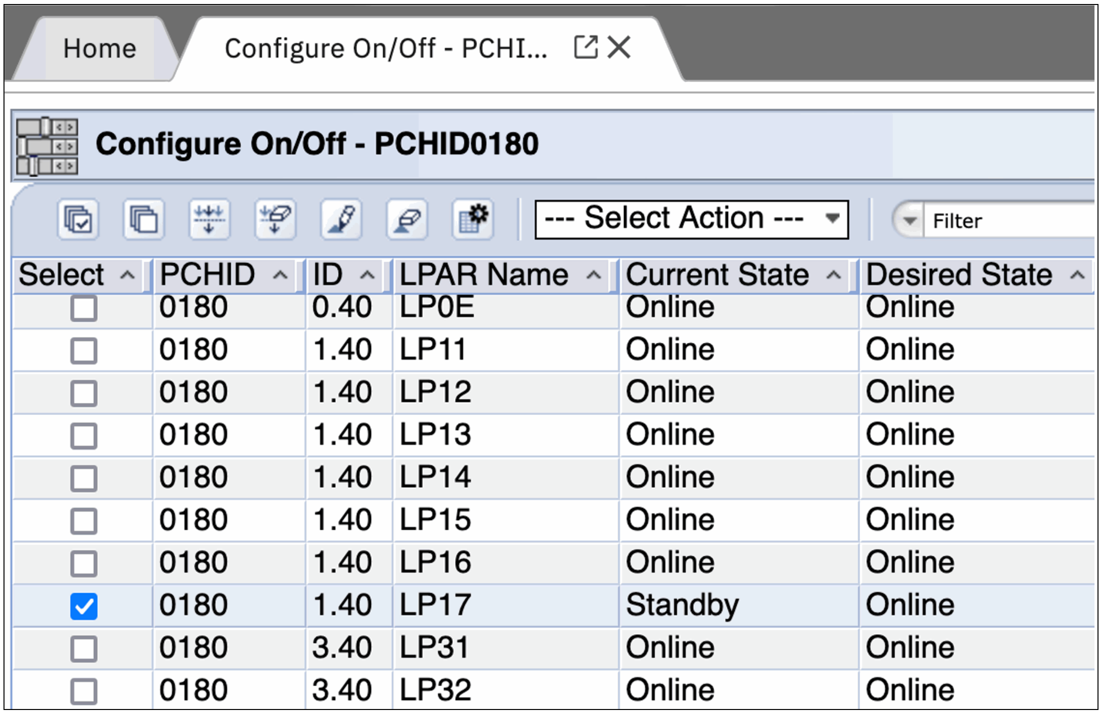


Figure 25 Updated panel

6. Click **OK** to close the completion notification. You might see a confirmation prompt if enabling this setting is disruptive.
7. With the CHPID online for the LPAR and the NPIV enabled, log out of the Service Element web interface (if applicable) by using the logout function in the upper right corner.
8. Repeat the previous steps to enable NPIV on all CHPIDs and LPARs where it is required.

> **Tip:** You can configure a CHPID offline from multiple LPARs at one time, enable NPIV on all of them, and then configure the CHPIDs back online all at once if you have many changes to make.

Finding an LPAR's NPIV WWPNs

After you enable NPIV, identify the corresponding NPIV WWPNs. Use the Hardware Management Console (HMC) and perform the following steps to identify the WWPNs.

1. Log on to the HMC (Figure 26 on page 20).

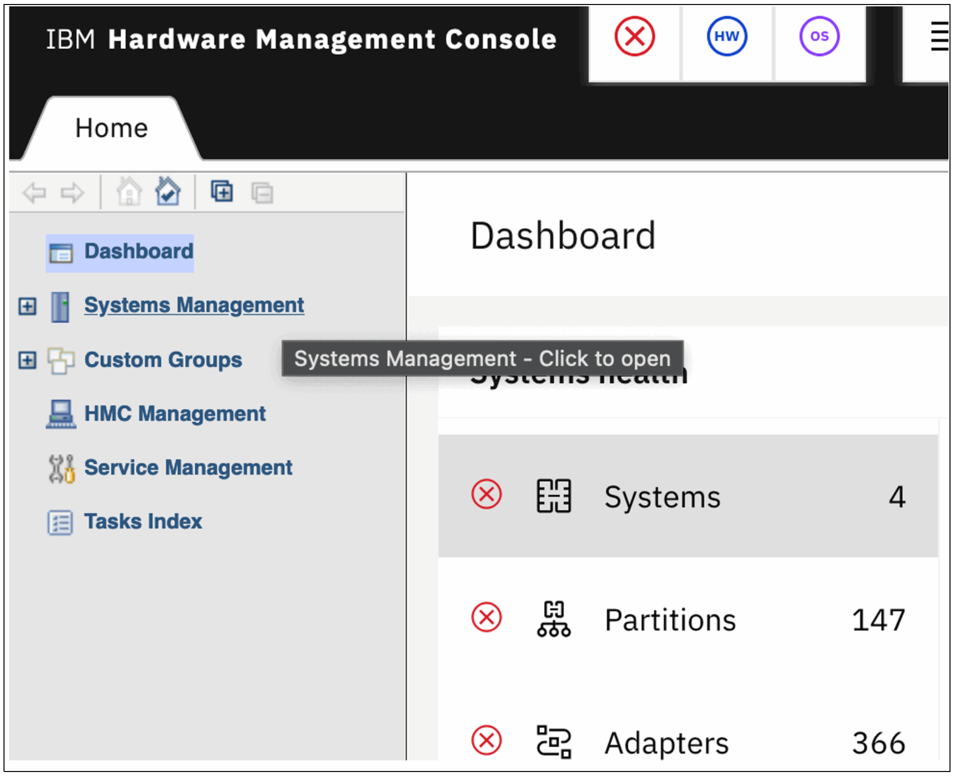


Figure 26 Hardware Management Console

2. Open the **Systems Management** tree and Select the **CPC** you want to examine (Figure 27).

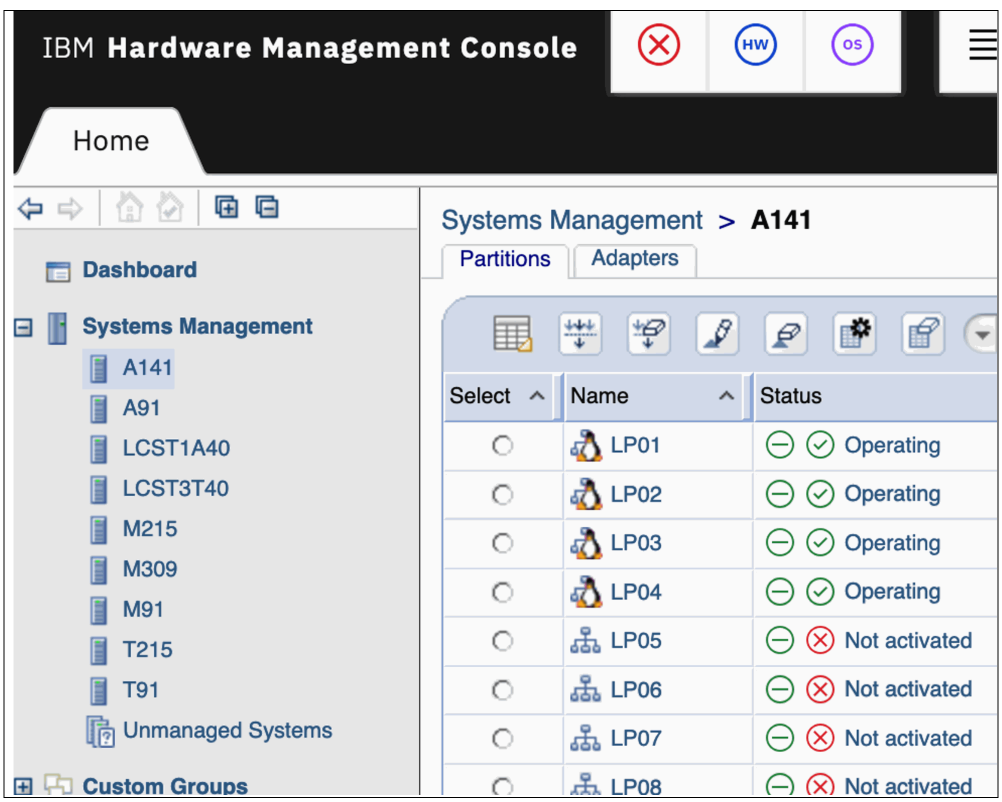


Figure 27 Select the CPC that you want to examine

3. Open the **Configuration** menu in the bottom center section, and select **FCP Configuration** (Figure 28).

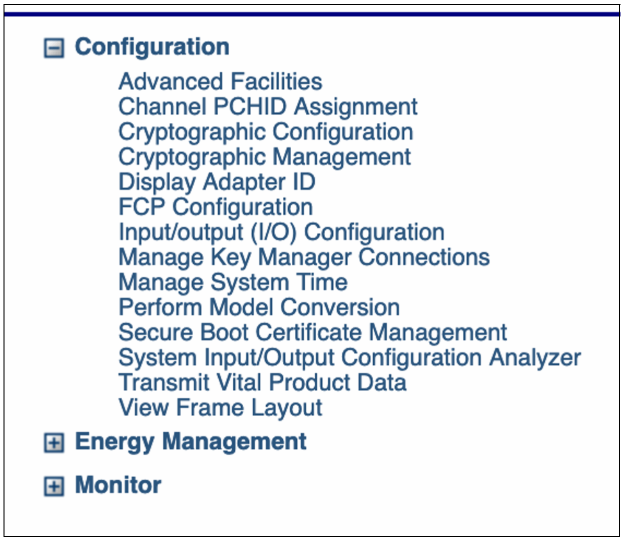


Figure 28 Select FCP Configuration

4. When the FCP Configuration tab opens, leave the option set to **Display all NPIV port names that are currently assigned to FCP subchannels…** and click **OK** (Figure 29).

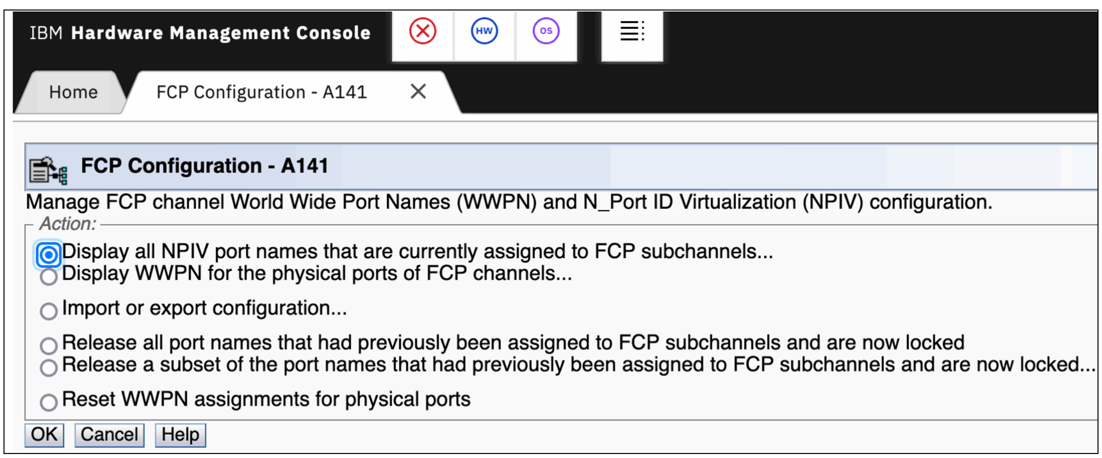


Figure 29 FCP Configuration tab

5. In the Display FCP NPIV Port Names menu, select **Display all assigned ports for an LPAR** and click **OK** (Figure 30 on page 22).

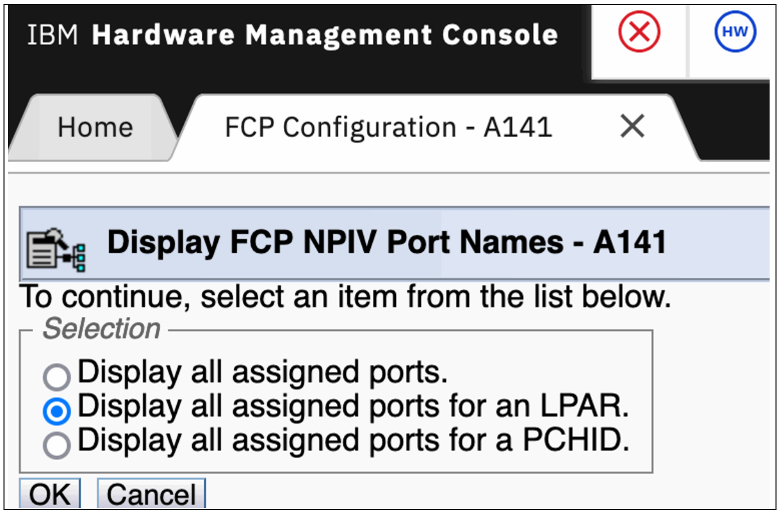


Figure 30 Set the option to Display all assigned ports for an LPAR

6. In the LPAR selection menus, select the LPAR for which you want to view the NPIV addresses. From the list of LPARs, select the wanted entry and click **OK**.

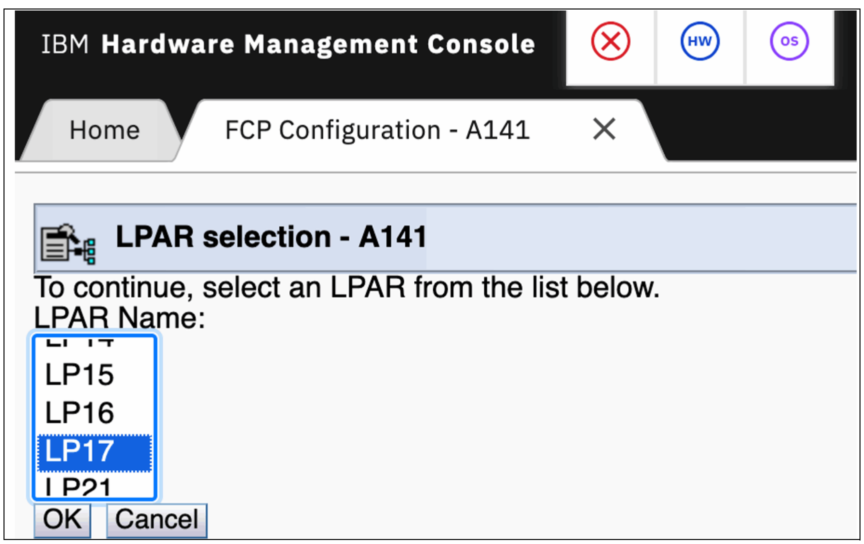
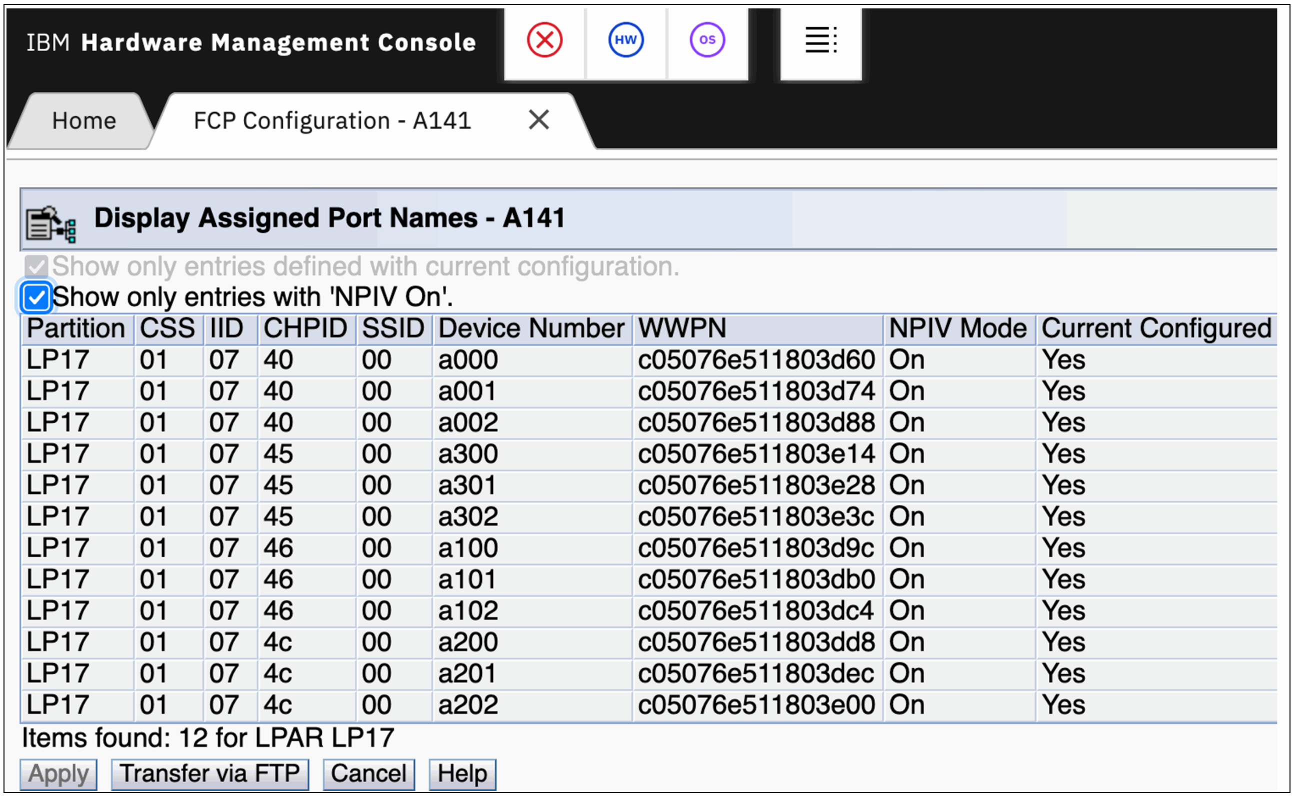


Figure 31 Select the LPAR that you want to view the NPIV addresses for

The device listing for your selected LPAR opens (Figure 32 on page 23).



Partition	CSS	IID	CHPID	SSID	Device Number	WWPN	NPIV Mode	Current Configured
LP17	01	07	40	00	a000	c05076e511803d60	On	Yes
LP17	01	07	40	00	a001	c05076e511803d74	On	Yes
LP17	01	07	40	00	a002	c05076e511803d88	On	Yes
LP17	01	07	45	00	a300	c05076e511803e14	On	Yes
LP17	01	07	45	00	a301	c05076e511803e28	On	Yes
LP17	01	07	45	00	a302	c05076e511803e3c	On	Yes
LP17	01	07	46	00	a100	c05076e511803d9c	On	Yes
LP17	01	07	46	00	a101	c05076e511803db0	On	Yes
LP17	01	07	46	00	a102	c05076e511803dc4	On	Yes
LP17	01	07	4c	00	a200	c05076e511803dd8	On	Yes
LP17	01	07	4c	00	a201	c05076e511803dec	On	Yes
LP17	01	07	4c	00	a202	c05076e511803e00	On	Yes

Figure 32 The device listing for your selected LPAR

Each device number on each CHPID has a unique WWPN, so the SAN and storage devices can differentiate between virtualized systems that are sharing the same physical host adapter. Without NPIV, devices a000, a001, and a002 within LPAR LP17 are indistinguishable to the SAN, granting them all access to the same SAN resources. Also, without NPIV, device a000 on LP17 is identical to device a000 on any other LPAR.

> **Tip:** For later use, save this data to a text file. Screenshots can be helpful, but they do not facilitate easy text extraction.

The zoning and storage example uses the following devices:

- LP17 a000 c05076e511803d60
- LP17 a100 c05076e511803d9c
- LP17 a200 c05076e511803dd8
- LP17 a300 c05076e511803e14

7. Click the **X** on the FCP Configuration tab to close this pane. Continue to click **X** on the tab until it closes to back all the way out of FCP Configuration.

   If your CHPIDs are not enabled for NPIV in this LPAR, you see an empty screen similar to Figure 33 on page 24.

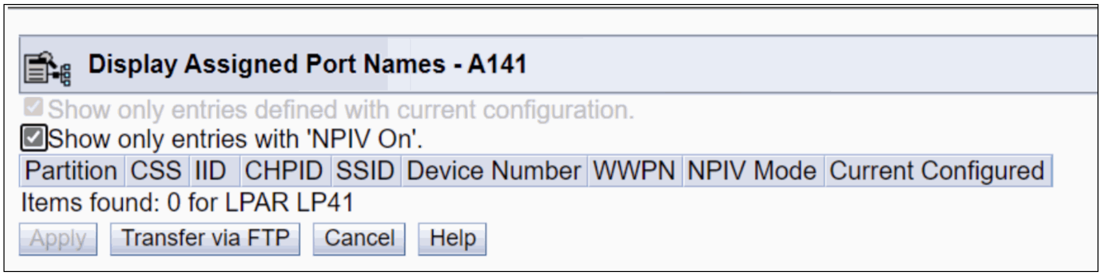


Figure 33 CHPIDs are not enabled for NPIV in this LPAR

8. Click **Cancel** to close the window.

Finding the correct WWPNs in a FlashSystem storage device

As storage systems offer more functions, their internal complexity grows to ensure compatibility with an earlier version. IBM FlashSystem devices gained numerous capabilities over time, and NPIV implementation added to the complexity. It is important to distinguish this storage subsystem NPIV implementation from the server-side NPIV discussed in “N_Port ID virtualization” on page 5.

IBM FlashSystem uses NPIV-enabled SAN ports for management isolation across various protocols, including SCSI, NVMe over Fibre Channel, and Metro/Global Mirror replication and for hardware failover high availability (HA).

An IBM FS9500 FlashSystem has a port configuration that looks like Figure 34, if it is fully populated with FCP adapters.

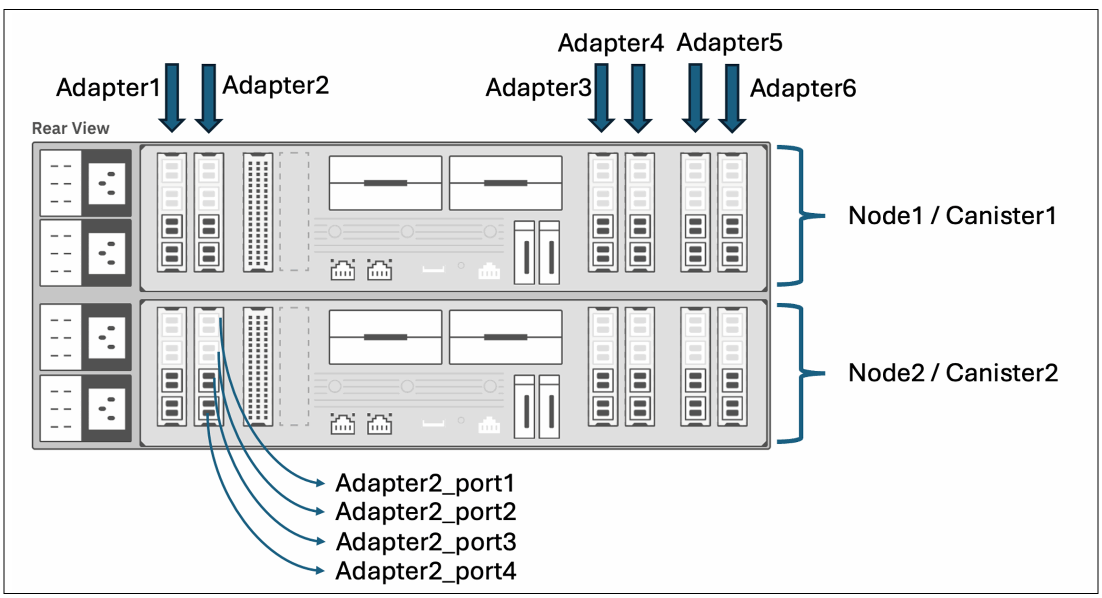


Figure 34 IBM FS9500 FlashSystem

Each IBM FlashSystem 9500 node consists of an upper and a lower node. Each node contains six I/O adapters, numbered 1 through 6 from left to right. Each adapter has multiple ports, numbered sequentially from top to bottom, starting with port 1.

Note: Other IBM FlashSystem models can have different physical orientations. For example, the lower node might be inverted, which results in reversed port numbering relative to the upper node. Consult the specific product documentation for each model to determine the correct port enumeration.

Identifying a specific port on the FS9500 requires all three components: Node, Adapter, and Port. Within the FS9500 management interface, FCP ports are assigned sequential port IDs within each node. This enumeration begins with adapter 1, port 1 (upper left), proceeds down that adapter, then moves to adapter 2, port 1, and continues down that adapter, and so forth, until the bottom port of adapter 6 is reached. Each node on the systems has 24 ports (6 adapters * 4 ports/adapter).

Figure 34 on page 24 of the web management interface highlights active, connected ports in a darker shade. The output in Figure 35 from the FlashSystem command-line interface (CLI), was generated using **`lsportfc -filtervalue "type=fc:status=active"`**. The command identifies the active FCP ports on the FS9500. Correlate the `Node:Adapter:Port` designations with the sequential port IDs to understand their mapping (Figure 35).

port_id	node	WWPN	nportid	status	adapter_location	adapter_port_id
3	node1	50050768131345AD	01C000	active	1	3
4	node1	50050768131445AD	E3D040	active	1	4
7	node1	50050768132345AD	9D0000	active	2	3
8	node1	50050768132445AD	E3D840	active	2	4
11	node1	50050768135345AD	01C100	active	5	3
12	node1	50050768135445AD	E5A3C0	active	5	4
15	node1	50050768136345AD	9D0020	active	6	3
16	node1	50050768136445AD	E3D800	active	6	4
19	node1	50050768137345AD	01C200	active	7	3
20	node1	50050768137445AD	E5A380	active	7	4
23	node1	50050768138345AD	9D0040	active	8	3
24	node1	50050768138445AD	E5A0C0	active	8	4
3	node2	50050768131345AE	12040	active	1	3
4	node2	50050768131445AE	E3D000	active	1	4
7	node2	50050768132345AE	9D0060	active	2	3
8	node2	50050768132445AE	E3BBC0	active	2	4
11	node2	50050768135345AE	12140	active	5	3
12	node2	50050768135445AE	E5A340	active	5	4
15	node2	50050768136345AE	9D0080	active	6	3
16	node2	50050768136445AE	E3BB80	active	6	4
19	node2	50050768137345AE	12240	active	7	3
20	node2	50050768137445AE	E5A300	active	7	4
23	node2	50050768138345AE	9D00A0	active	8	3
24	node2	50050768138445AE	E5A080	active	8	4

Figure 35 Port mappings

The WWPNs listed in Figure 35 are the physical WWPNs on those FCP ports, and they are *not* for host I/O. Do not zone any of the physical WWPNs to a host for running any I/O.

Note: In current FlashSystems firmware, the physical WWPNs are only for the node failover communications zones and for the Global and Metro Mirror zones.

With NPIV enabled, each FCP port on an IBM FlashSystem storage unit presents three WWPNs to the SAN, as illustrated in Figure 36. The trimmed output of **`lstargetportfc -filtervalue "port_id=3:owning_node_id=1"`** shows all logical target ports on port 3 of port 1, which is the third port down on adapter 1 on node 1.

id	WWPN	WWNN	port_id	node_id	nportid	host_io_perm	virt	prot
7	50050768131345AD	50050768130045AD	3	1	01C000	no	no	scsi
8	50050768131745AD	50050768130045AD	3	1	01C001	yes	yes	scsi
9	50050768131B45AD	50050768130045AD	3	1	01C002	yes	yes	nvme

Figure 36 Logical target ports

The table in Figure 36 lists three WWPNs associated with a single physical port that is logged into the SAN switch. The physical WWPN, `50050768131345AD` (previously shown in Figure 35 on page 25), has a `host_io_permitted` status of `no`. Additionally, two NPIV-enabled ports are logged in to the fabric: one for FCP SCSI protocol I/O (`50050768131745AD`) and the other for NVMe over Fibre Channel I/O (`50050768131B45AD`).

Do not zone the physical port WWPN to a host; it is not authorized for host I/O. Similarly, ensure that neither virtual WWPN is zoned to a host unless that host is using the corresponding protocol. Specifically, do not zone an FCP SCSI host to the NVMe port, and do not zone an NVMe over Fibre Channel host to an FCP SCSI port.

Tip: Although zoning a host to inappropriate ports on the FlashSystem might not immediately generate errors, it can complicate subsequent troubleshooting. To facilitate future diagnostics, ensure that hosts are zoned only to the appropriate ports on the storage controller.

Figure 37 on page 27 shows the output of **`lstargetportfc -filtervalue "virtualized=yes:protocol=scsi",`** which lists all the valid NPIV target ports for SCSI protocol hosts.

For the example configuration, only the WWPNs that are listed in Figure 37 are valid for zoning with the LinuxONE hosts. Examining the `nportid` column reveals that the device has connections to three separate switches. Specifically, Port 4 on all adapters connects to one switch; Port 3 on adapters 1, 3, and 5 connects to a second switch; and Port 3 on adapters 2, 4, and 6 connects to a third.

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
2	50050768131545AD	50050768130045AD	1	1	0	yes	scsi
5	50050768131645AD	50050768130045AD	2	1	0	yes	scsi
8	50050768131745AD	50050768130045AD	3	1	01C001	yes	scsi
11	50050768131845AD	50050768130045AD	4	1	E3D041	yes	scsi
14	50050768132545AD	50050768130045AD	5	1	0	yes	scsi
17	50050768132645AD	50050768130045AD	6	1	0	yes	scsi
20	50050768132745AD	50050768130045AD	7	1	9D0001	yes	scsi
23	50050768132845AD	50050768130045AD	8	1	E3D841	yes	scsi
26	50050768135545AD	50050768130045AD	9	1	0	yes	scsi
29	50050768135645AD	50050768130045AD	10	1	0	yes	scsi
32	50050768135745AD	50050768130045AD	11	1	01C101	yes	scsi
35	50050768135845AD	50050768130045AD	12	1	E5A3C1	yes	scsi
38	50050768136545AD	50050768130045AD	13	1	0	yes	scsi
41	50050768136645AD	50050768130045AD	14	1	0	yes	scsi
44	50050768136745AD	50050768130045AD	15	1	9D0021	yes	scsi
47	50050768136845AD	50050768130045AD	16	1	E3D801	yes	scsi
50	50050768137545AD	50050768130045AD	17	1	0	yes	scsi
53	50050768137645AD	50050768130045AD	18	1	0	yes	scsi
56	50050768137745AD	50050768130045AD	19	1	01C201	yes	scsi
59	50050768137845AD	50050768130045AD	20	1	E5A381	yes	scsi
62	50050768138545AD	50050768130045AD	21	1	0	yes	scsi
65	50050768138645AD	50050768130045AD	22	1	0	yes	scsi
68	50050768138745AD	50050768130045AD	23	1	9D0041	yes	scsi
71	50050768138845AD	50050768130045AD	24	1	E5A0C1	yes	scsi
74	50050768131545AE	50050768130045AE	1	2	0	yes	scsi
77	50050768131645AE	50050768130045AE	2	2	0	yes	scsi
80	50050768131745AE	50050768130045AE	3	2	012041	yes	scsi
83	50050768131845AE	50050768130045AE	4	2	E3D001	yes	scsi
86	50050768132545AE	50050768130045AE	5	2	0	yes	scsi
89	50050768132645AE	50050768130045AE	6	2	0	yes	scsi
92	50050768132745AE	50050768130045AE	7	2	9D0061	yes	scsi
95	50050768132845AE	50050768130045AE	8	2	E3BBC1	yes	scsi
98	50050768135545AE	50050768130045AE	9	2	0	yes	scsi
101	50050768135645AE	50050768130045AE	10	2	0	yes	scsi
104	50050768135745AE	50050768130045AE	11	2	012141	yes	scsi
107	50050768135845AE	50050768130045AE	12	2	E5A341	yes	scsi
110	50050768136545AE	50050768130045AE	13	2	0	yes	scsi
113	50050768136645AE	50050768130045AE	14	2	0	yes	scsi
116	50050768136745AE	50050768130045AE	15	2	9D0081	yes	scsi
119	50050768136845AE	50050768130045AE	16	2	E3BB81	yes	scsi
122	50050768137545AE	50050768130045AE	17	2	0	yes	scsi
125	50050768137645AE	50050768130045AE	18	2	0	yes	scsi
126	50050768137745AE	50050768130045AE	19	2	012241	yes	scsi
131	50050768137845AE	50050768130045AE	20	2	E5A301	yes	scsi
134	50050768138545AE	50050768130045AE	21	2	0	yes	scsi
137	50050768138645AE	50050768130045AE	22	2	0	yes	scsi
140	50050768138745AE	50050768130045AE	23	2	9D00A1	yes	scsi
143	50050768138845AE	50050768130045AE	24	2	E5A081	yes	scsi

Figure 37 Valid NPIV target ports for SCSI protocol hosts

The following examples focus solely on the two switches connected to Port 3. So, the relevant ports in Figure 37 on page 27 are listed in tables in Figure 38 and Figure 39, one table for each of these switches.

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
8	50050768131745AD	50050768130045AD	3	1	01C001	yes	scsi
32	50050768135745AD	50050768130045AD	11	1	01C101	yes	scsi
56	50050768137745AD	50050768130045AD	19	1	01C201	yes	scsi
80	50050768131745AE	50050768130045AE	3	2	012041	yes	scsi
104	50050768135745AE	50050768130045AE	11	2	012141	yes	scsi
126	50050768137745AE	50050768130045AE	19	2	012241	yes	scsi

Figure 38 WWPNs to use for FCP target ports - Switch one

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
20	50050768132745AD	50050768130045AD	7	1	9D0001	yes	scsi
44	50050768136745AD	50050768130045AD	15	1	9D0021	yes	scsi
68	50050768138745AD	50050768130045AD	23	1	9D0041	yes	scsi
92	50050768132745AE	50050768130045AE	7	2	9D0061	yes	scsi
116	50050768136745AE	50050768130045AE	15	2	9D0081	yes	scsi
140	50050768138745AE	50050768130045AE	23	2	9D00A1	yes	scsi

Figure 39 WWPNs to use for FCP target ports - Switch two

The WWPNs presented in Figure 38 and Figure 39 constitute the exclusive set of ports that are employed for FCP target port zoning within the subsequent section. The construction of these tables is based on the previously generated output from the command **`lstargetportfc -filtervalue "virtualized=yes:protocol=scsi"`**, which lists all NPIV SCSI ports. The output of the command is corroborated with documentation that details the physical fiber link connections, and is then verified against the nportid values, which serve as indicators of connections to distinct SAN fabrics.

For the configuration of FlashSystem chassis internal connectivity zones, it is also necessary to obtain the physical WWPNs of all connected ports on the two target switches. The output in Figure 40 and Figure 41 on page 29 was generated by using the command **`lstargetportfc -filtervalue "virtualized=no"`** and is arranged to categorize the results by switch connection.

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
7	50050768131345AD	50050768130045AD	3	1	01C000	no	scsi
31	50050768135345AD	50050768130045AD	11	1	01C100	no	scsi
55	50050768137345AD	50050768130045AD	19	1	01C200	no	scsi
79	50050768131345AE	50050768130045AE	3	2	012040	no	scsi
103	50050768135345AE	50050768130045AE	11	2	012140	no	scsi
127	50050768137345AE	50050768130045AE	19	2	012240	no	scsi

Figure 40 Physical WWPNs of all connected ports - Switch one

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
19	50050768132345AD	50050768130045AD	7	1	9D0000	no	scsi
43	50050768136345AD	50050768130045AD	15	1	9D0020	no	scsi
67	50050768138345AD	50050768130045AD	23	1	9D0040	no	scsi
91	50050768132345AE	50050768130045AE	7	2	9D0060	no	scsi
115	50050768136345AE	50050768130045AE	15	2	9D0080	no	scsi
139	50050768138345AE	50050768130045AE	23	2	9D00A0	no	scsi

Figure 41 Physical WWPNs of all connected ports - Switch two

> **Important:** As stated previously, do not use the physical WWPNs listed in these tables for connections from remote Fibre Channel hosts. Their purpose is limited to traffic associated with port-to-port failover within the chassis or inter-chassis communication for Peer-to-Peer Remote Copy, Metro Mirror, and Global Mirror replication.

Creating SAN zones to permit host ports to communicate with storage ports

The host-side WWPNs for LP17, with the corresponding switch connections added that are based on infrastructure documentation are shown in Example 2.

Example 2 Host-side WWPNs for LP17

```
lpardevicewwpn              switch
LP17 a000 c05076e511803d60 sw one
LP17 a100 c05076e511803d9c sw two
LP17 a200 c05076e511803dd8 sw one
LP17 a300 c05076e511803e14 sw two
```

Example 3 lists the WWPNs for LP16.

Example 3 Host-side WWPNs for LP16

```
lpardevicewwpn              switch
LP16 a000 c05076e511803d5c sw one
LP16 a100 c05076e511803d98 sw two
LP16 a200 c05076e511803dd4 sw one
LP16 a300 c05076e511803e10 sw two
```

Also, the ports on the FS9500 are connected and available (Figure 42 and Figure 43 on page 30).

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
8	50050768131745AD	50050768130045AD	3	1	01C001	yes	scsi
32	50050768135745AD	50050768130045AD	11	1	01C101	yes	scsi
56	50050768137745AD	50050768130045AD	19	1	01C201	yes	scsi
80	50050768131745AE	50050768130045AE	3	2	012041	yes	scsi
104	50050768135745AE	50050768130045AE	11	2	012141	yes	scsi
126	50050768137745AE	50050768130045AE	19	2	012241	yes	scsi

Figure 42 Ports on the FS9500 - Switch one

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
20	50050768132745AD	50050768130045AD	7	1	9D0001	yes	scsi
44	50050768136745AD	50050768130045AD	15	1	9D0021	yes	scsi
68	50050768138745AD	50050768130045AD	23	1	9D0041	yes	scsi
92	50050768132745AE	50050768130045AE	7	2	9D0061	yes	scsi
116	50050768136745AE	50050768130045AE	15	2	9D0081	yes	scsi
140	50050768138745AE	50050768130045AE	23	2	9D00A1	yes	scsi

Figure 43 Ports on the FS9500 - Switch two

The overall connectivity is illustrated in Figure 44.

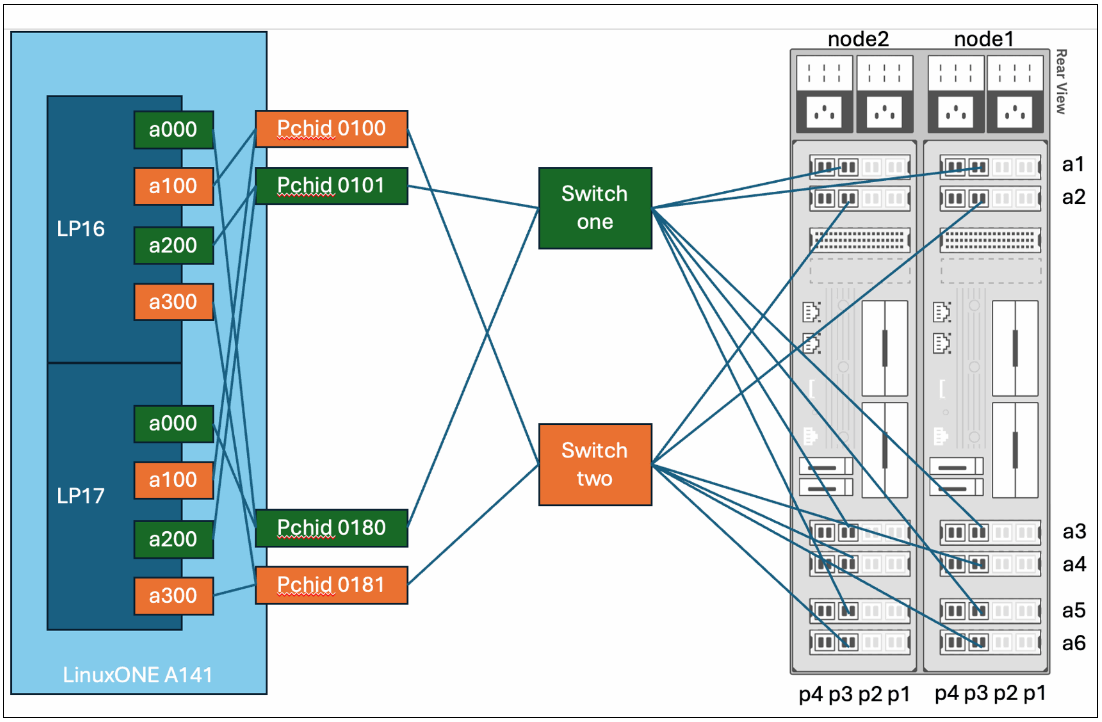


Figure 44 Overall connectivity diagram

Based on these connections, you can create zones on switch one similar to the ones in Example 4 and Example 5.

Example 4 Creating zones for FS9500 - switch one

```
fs9500_a141_path0_switch1
  host LP16 a000
  host LP17 a000
  target node1:adapter3:port3 - id 32
  target node2:adapter3:port3 - id 104
```

Example 5 Creating zones for FS9500 - switch one

```
fs9500_a141_path2_switch1
  host LP16 a200
  host LP17 a200
  target node1:adapter5:port3 - id 56
  target node2:adapter5:port3 - id 126
```

Figure 45 illustrates the connectivity diagram based on the configuration in Example 4 on page 30.

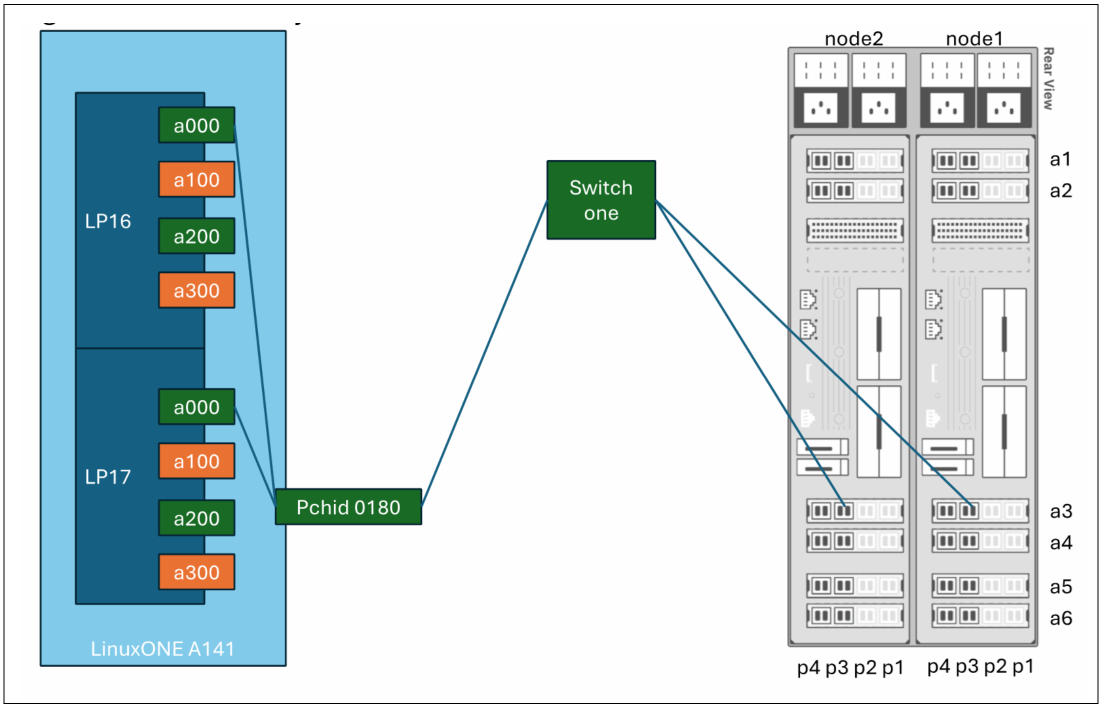


Figure 45 Connectivity diagram resulting from the configuration in Example 4

Figure 46 illustrates the connectivity diagram based on the configuration in Example 5 on page 30.

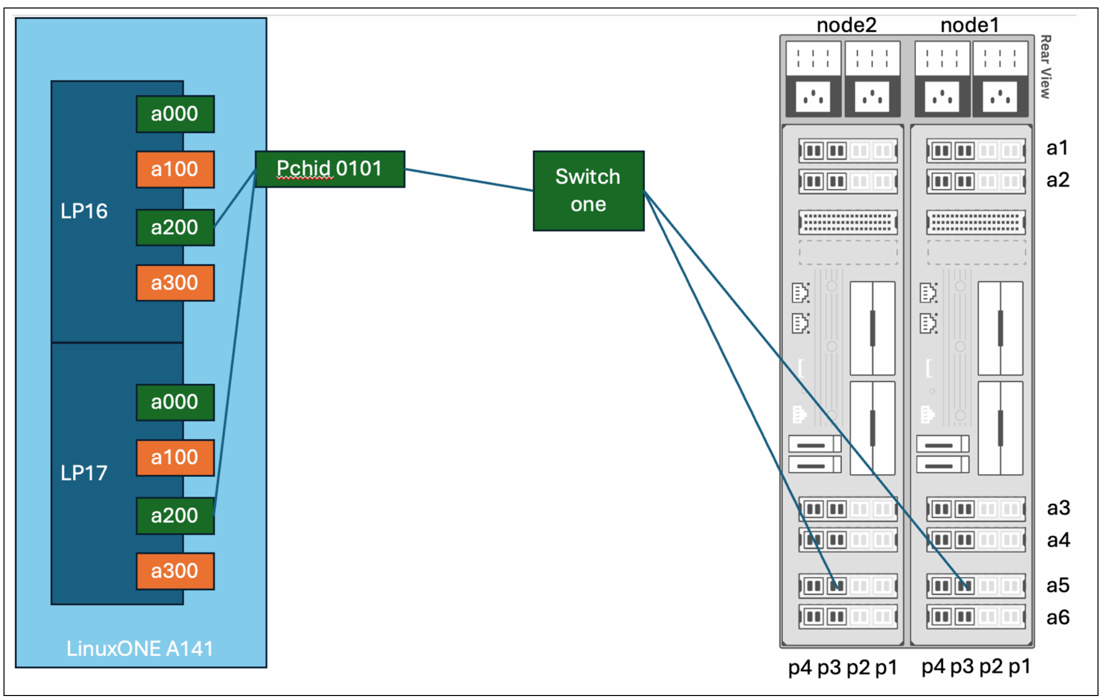


Figure 46 Connectivity diagram resulting from the configuration in Example 5

The zones that are created on switch two look like Example 6 and Example 7.

Example 6 Creating zones for FS9500 - switch two

```
fs9500_a141_path1_switch2
  host LP16 a100
  host LP17 a100
  target node1:adapter2:port3 - id 20
  target node2:adapter2:port3 - id 92
```

Example 7 Creating zones for FS9500 - switch two

```
fs9500_a141_path3_switch2
  host LP16 a300
  host LP17 a300
  target node1:adapter4:port3 - id 44
  target node2:adapter4:port3 - id 116
```

Figure 47 shows the connectivity diagram based on the configuration in Example 6.

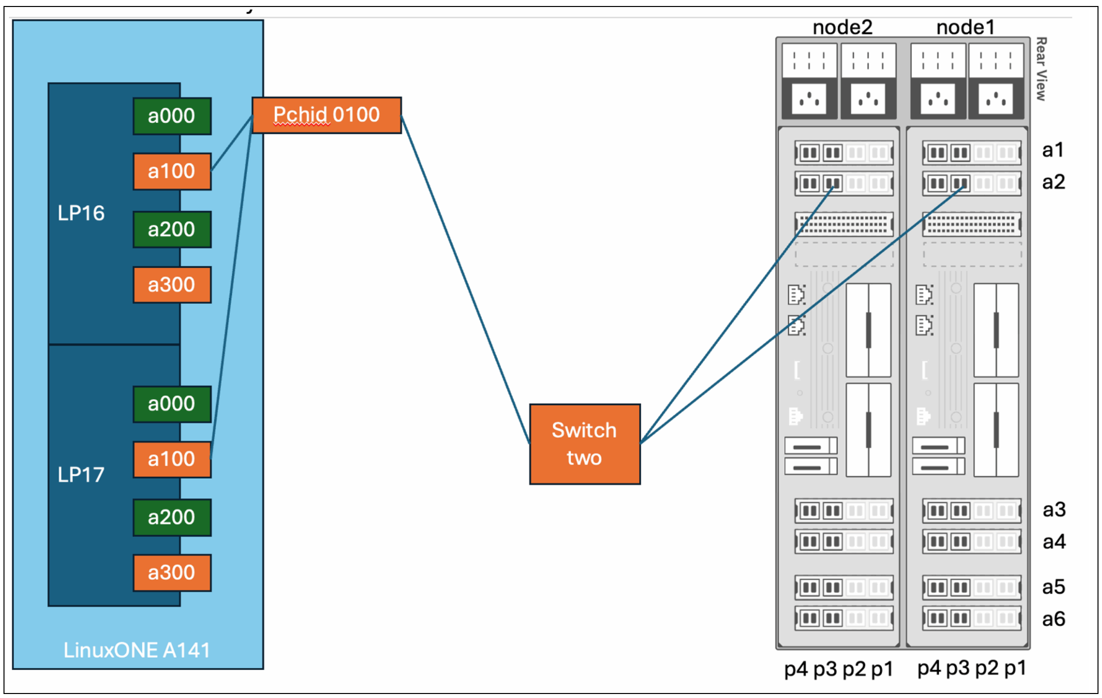


Figure 47 Connectivity diagram resulting from the configuration in Example 6

Figure 48 on page 33 shows the connectivity diagram based on the configuration in Example 7.

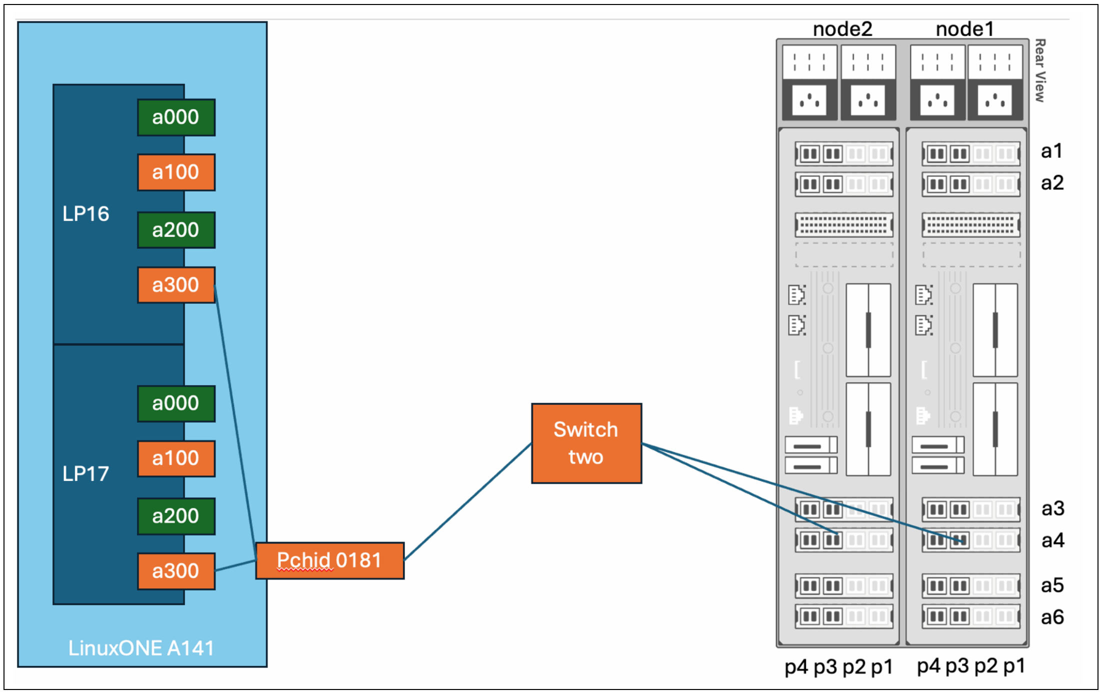


Figure 48 Connectivity diagram resulting from the configuration in Example 7

In the example configuration, all ports on adapters 1 and 6 in both FlashSystem nodes are dedicated to node-to-node failover zones and Metro/Global Mirror communication. Host connections are exclusively configured to use ports on adapters 2, 3, 4, and 5. Odd-numbered adapters connect to switch one, and even-numbered adapters connect to switch two.

Each zone includes two target ports on the storage unit, which is on the same adapter and port number on both nodes. For example, node 1, adapter 4, port 3, and node 2, adapter 4, port 3, are paired in a zone. This configuration is intentional. The FlashSystem firmware can relocate NPIV WWPNs between nodes during certain failure or service events. This relocation occurs on a corresponding adapter-port basis. Therefore, during a failover, the NPIV SCSI WWPN for adapter 2, port 3 on node 1 is relocated to adapter 2, port 3 on node 2. It is not relocated to a different port on adapter 2 or to a different adapter on node 2.

The switches support an advanced zoning feature called Peer Zones on Brocade devices and Smart Zones on Cisco devices. By using this feature, administrators can classify zone members into two distinct categories. Although the specific terminology varies between vendors, the fundamental concept involves two member types, designated here as Type 1 and Type 2. When adding a member to a zone, it is assigned as either Type 1 or Type 2. Members of Type 1 are aware of all Type 2 members, and Type 2 members are aware of all Type 1 members. Critically, Type 1 members are not aware of other Type 1 members.

Use Peer Zones or Smart Zones if they are supported on the switch. These advanced zoning capabilities can simplify the management of complex storage networks and reduce the overall number of zones required.

In the absence of these advanced zoning features, Single Initiator Zoning represents the next most effective approach. In the previous switch configuration, this method resulted in the management of thousands of zones. Single Initiator Zoning involves creating a zone containing one host initiator and a set of target adapters. This zone is then replicated for each host requiring access to those same target adapters. Therefore, in a SAN with 1,000 hosts, each with four initiators communicating with the same set of eight target ports, 4,000 zones are necessary. In contrast, by using Peer Zones or Smart Zones, you can achieve the same scenario with as few as four zones with each zone containing 1,000 initiators and two targets.

> **Important:** On IBM LinuxONE and IBM Z, the host port NPIV WWPN does not register with the SAN fabric until the corresponding LPAR is activated, an operating system is loaded, and the operating system subsequently configures the device online. The storage administrator requires the necessary host-side WWPNs to configure zones. These WWPNs must be provided to the storage administrator.
>
> NPIV storage target WWPNs on an IBM FlashSystem register with the network shortly after the storage system is powered on and are therefore visible within the fabric. However, provide the storage administrator with explicit direction and specify which of the available NPIV WWPNs you are using.

The actual zone configuration that is implemented on Brocade switch one for these hosts is shown in Example 8.

Example 8 Zone configuration implemented on Brocade switch one

```
lpardevicewwpn           switch
LP17 a000 c05076e511803d60 sw one
LP17 a200 c05076e511803dd8 sw one
LP16 a000 c05076e511803d5c sw one
LP16 a200 c05076e511803dd4 sw one
```

Connect the hosts in Example 8 to the storage ports in Figure 49.

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
32	50050768135745AD	50050768130045AD	11	1	01C101	yes	scsi
56	50050768137745AD	50050768130045AD	19	1	01C201	yes	scsi
104	50050768135745AE	50050768130045AE	11	2	012141	yes	scsi
126	50050768137745AE	50050768130045AE	19	2	012241	yes	scsi

Figure 49 Storage ports

Before configuring zones, create aliases to enhance manageability and clarity (Example 9).

Example 9 Creating aliases

```
alicreate "a141_lp17_a000_hostname0","c0:50:76:e5:11:80:3d:60"
alicreate "a141_lp17_a200_hostname0","c0:50:76:e5:11:80:3d:d8"
alicreate "a141_lp16_a000_hostname1","c0:50:76:e5:11:80:3d:5c"
alicreate "a141_lp16_a200_hostname1","c0:50:76:e5:11:80:3d:d4"
alicreate "fs95fp1_node1_a3p3_npiv","50:05:07:68:13:57:45:ad"
alicreate "fs95fp1_node2_a3p3_npiv","50:05:07:68:13:57:45:ae"
alicreate "fs95fp1_node1_a5p3_npiv","50:05:07:68:13:77:45:ad"
alicreate "fs95fp1_node2_a5p3_npiv","50:05:07:68:13:77:45:ae"
```

Define the peer zones. Storage target WWPNs are designated as `-principal` members, and host initiator WWPNs are designated as `-member` members.

It is important to note the selective approach to port connectivity. Although the Linux multipath driver can handle a maximum of eight paths to a LUN, adding more paths beyond this threshold increases SAN usage without proportionally improving performance or resilience. A full mapping of all target ports to all initiator ports using Smart Zone or Peer zones creates 16 paths, which is suboptimal. The example configuration establishes a mapping of only two target ports to each LPAR's initiator on a shared adapter (Example 10).

Example 10 Defining the peer zones

```
zonecreate --peerzone "a141hosts_fs95fp1_port_33" \
-principal "fs95fp1_node1_a3p3_npiv;fs95fp1_node2_a3p3_npiv" \
-members "a141_lp17_a000_hostname0;a141_lp16_a000_hostname1"

zonecreate --peerzone "a141hosts_fs95fp1_port_53" \
-principal "fs95fp1_node1_a5p3_npiv;fs95fp1_node2_a5p3_npiv" \
-members"a141_lp17_a200_hostname0;a141_lp16_a200_hostname1"
```

Also, within the FlashSystem storage controller itself, zone the ports to support various service and failover recovery scenarios.

Because these ports are dedicated for internal controller communication and are not used by external systems, aliases are not required for these zones.

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
7	50050768131345AD	50050768130045AD	3	1	01C000	no	scsi
31	50050768135345AD	50050768130045AD	11	1	01C100	no	scsi
55	50050768137345AD	50050768130045AD	19	1	01C200	no	scsi
79	50050768131345AE	50050768130045AE	3	2	012040	no	scsi
103	50050768135345AE	50050768130045AE	11	2	012140	no	scsi
127	50050768137345AE	50050768130045AE	19	2	012240	no	scsi

Figure 50 Configuring and the zone is:

The zone configuration is shown in Example 11.

Example 11 Zone configuration

```
zonecreate "fs95fp1_local_only" \
"50050768131345AD; 50050768135345AD;50050768137345AD;50050768131345AE; \
  50050768135345AE; 50050768137345AE"
```

To reiterate, these physical WWPNs for the FlashSystem storage ports are exclusively for interstorage communication (either within the chassis or between FlashSystem systems). They must never be zoned to host systems for host I/O.

Add the newly created zones to the zoneset. Save the changes and enable the new zoneset (Example 12).

Example 12 Add zones to zoneset

```
cfgadd "ZONE14_0","a141hosts_fs95fp1_port_33;a141hosts_fs95fp1_port_53, \
  fs95fp1_local_only"
cfgsave
cfgenable "ZONE14_0"
```

Similarly, on the other switch, which is a Cisco, zone the hosts in Example 13 on page 36 to the storage ports shown in Figure 51 on page 36.

Example 13 Create zones on the Cisco switch

```
LP17 a100 c05076e511803d9c sw two
LP17 a300 c05076e511803e14 sw two
LP16 a100 c05076e511803d98 sw two
LP16 a300 c05076e511803e10 sw two
```

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
20	50050768132745AD	50050768130045AD	7	1	9D0001	yes	scsi
44	50050768136745AD	50050768130045AD	15	1	9D0021	yes	scsi
92	50050768132745AE	50050768130045AE	7	2	9D0061	yes	scsi
116	50050768136745AE	50050768130045AE	15	2	9D0081	yes	scsi

Figure 51 Storage ports on the Cisco switch

Create aliases to improve readability (Example 14).

Example 14 Creating aliases

```
config t
device-alias database
device-alias name a141_lp17_a100_hostname0 pwwn C0:50:76:E5:11:80:3D:9C
device-alias name a141_lp17_a300_hostname0 pwwn C0:50:76:E5:11:80:3E:14
device-alias name a141_lp16_a100_hostname1 pwwn C0:50:76:E5:11:80:3D:98
device-alias name a141_lp16_a300_hostname1 pwwn C0:50:76:E5:11:80:3E:10
device-alias name fs95fp1_node1_a2p3_npiv pwwn 50:05:07:68:13:27:45:AD
device-alias name fs95fp1_node2_a2p3_npiv pwwn 50:05:07:68:13:27:45:AE
device-alias name fs95fp1_node1_a4p3_npiv pwwn 50:05:07:68:13:67:45:AD
device-alias name fs95fp1_node2_a4p3_npiv pwwn 50:05:07:68:13:67:45:AE
device-alias commit
```

Create the new zones. The `target` argument indicates these are storage target WWPNs, and the `init` argument indicates these are host initiator WWPNs (Example 15).

Example 15 Creating new zones

```
config t
zone name a141hosts_fs95sp1_port_23 vsan 15
member device-alias fs95fp1_node1_a2p3_npiv target
member device-alias fs95fp1_node2_a2p3_npiv target
member device-alias a141_lp17_a100_hostname0 init
member device-alias a141_lp16_a100_hostname1 init

zone name a141hosts_fs95sp1_port_43 vsan 15
member device-alias fs95fp1_node1_a4p3_npiv target
member device-alias fs95fp1_node2_a4p3_npiv target
member device-alias a141_lp17_a300_hostname0 init
member device-alias a141_lp16_a300_hostname1 init
```

As it was done on the other switch, create the local-only zone for intranode communication. Figure 52 shows the physical ports that are connected to this switch.

id	WWPN	WWNN	port_id	node	nportid	virtualized	protocol
19	50050768132345AD	50050768130045AD	7	1	9D0000	no	scsi
43	50050768136345AD	50050768130045AD	15	1	9D0020	no	scsi
67	50050768138345AD	50050768130045AD	23	1	9D0040	no	scsi
91	50050768132345AE	50050768130045AE	7	2	9D0060	no	scsi
115	50050768136345AE	50050768130045AE	15	2	9D0080	no	scsi
139	50050768138345AE	50050768130045AE	23	2	9D00A0	no	scsi

Figure 52 Physical ports plugged to switch two

Create a standard zone for these ports as shown in Example 16.

Example 16 Create a standard zone

```
zone name fs95fp1_local_only vsan 15
member pwwn 50050768132345AD
member pwwn 50050768136345AD
member pwwn 50050768138345AD
member pwwn 50050768132345AE
member pwwn 50050768136345AE
member pwwn 50050768138345AE
```

To reiterate again, the listed WWPNs are the physical WWPNs of the FlashSystem, and are only ever zoned to each other or other FlashSystem physical WWPNs. They are not for host I/O.

Lastly, add the new zones to a zoneset, activate the changes, and save them (Example 17).

Example 17 Add the new zones to a zoneset, activate the changes, and save them

```
zoneset name ZONE15_0 vsan 15
member a141hosts_fs95sp1_port_23
member a141hosts_fs95sp1_port_43
member fs95fp1_local_only
zoneset activate name ZONE15_0 vsan 15
end
copy running-config startup-config
```

As of this point, the zoning is completed for the NPIV host ports of the IBM LinuxONE or IBM Z to the FS9500 storage ports on the Brocade and Cisco switches.

Although the initial configuration includes only two hosts in these zones, Peer Zones and Smart Zones offer the significant benefit of simplified scaling. To provide additional hosts with access to the storage controller, you can add the NPIV WWPNs of the hosts to the pre-existing zones. This method also streamlines the identification of initiator ports that are communicating with a particular target port. Instead of searching across numerous zones to locate a specific WWPN and count the single initiator zones where it appears, you can locate the relevant zone that contains the storage WWPN and count the number of initiator WWPNs within that zone.

Creating host definitions and mapping volumes in the storage controller

The FS9500 offers both a web UI and a CLI. These examples use the CLI, as it provides a more robust foundation for automation.

Recall that the NPIV WWPNs of two LPARs are listed in Example 18 and Example 19.

Example 18 WWPNs for LP17

```
lpardevicewwpn
LP17 a000 c05076e511803d60
LP17 a200 c05076e511803dd8
LP16 a000 c05076e511803d5c
LP16 a200 c05076e511803dd4
```

Example 19 WWPNs for LP16

```
lpardevicewwpn
LP16 a000 c05076e511803d5c
LP16 a100 c05076e511803d98
LP16 a200 c05076e511803dd4
LP16 a300 c05076e511803e10
```

Host definitions must be created on the FS9500 storage controller before volumes can be mapped to host systems. This presents a potential bootstrapping challenge when the target systems boot from these volumes. The web UI offers a method for creating host definitions by presenting a list of previously unseen WWPNs that are connected to the storage system. Selecting these unknown WWPNs creates a new host definition, which enables volume mapping. However, if the target systems are not yet zoned to the storage or if they lack a running operating system, these WWPNs are not available for selection.

Because the zone creation process uses offline WWPNs, you can maintain consistency by using the FS9500 CLI. Use the **`mkhost`** command with the **`-force`** parameter because the FlashSystem code typically validates all provided WWPNs. The validation fails if the LPAR is not yet activated with an operating system that is loaded and running and the devices configured online (Example 20).

Example 20 mkhost command

```
mkhost -name A141_LP17_hostname0 -protocol fcscsi -fcwwpn \
c05076e511803d60:c05076e511803d9c:c05076e511803dd8:c05076e511803e14 -force
mkhost -name A141_LP16_hostname1 -protocol fcscsi -fcwwpn \
c05076e511803d5c:c05076e511803d98:c05076e511803dd4:c05076e511803e10 -force
```

Then, map an existing volume to each host definition:

```
mkvdiskhostmap -host A141_LP17_hostname0 <some_volume_id>
mkvdiskhostmap -host A141_LP16_hostname1 <some_other_volume_id>
```

This completes the basic storage configuration on the SAN storage controller for simple volume mappings (one volume per host). For clustered file systems such as IBM Storage Scale (formerly GPFS), mapping a volume or set of volumes to multiple hosts requires the **`-force`** option or the **`mkhostcluster`** command. These methods enable the FlashSystem to gracefully manage shared-volume access across multiple hosts.

Booting new systems from SAN volumes while simultaneously sharing a set of volumes across a host cluster requires careful planning and design. Do not permit multiple systems to access each others' boot volumes that use nonclustered file systems, such as ext4, xfs, or btrfs. Inadvertent concurrent access corrupts the data on these volumes, potentially rendering the operating system unusable if the affected volume is a boot volume.

To prevent shared access to boot volumes, you can use the **`-ignoreseedvolume`** parameter when you create the host cluster. However, for maximum assurance, it is recommended to employ multiple host adapters per channel, dedicating one set for operating system volume access and another set for cluster volume access.

For example, on LP17 you can continue to use the ports in Example 21 for access to the boot volume.

Example 21 Ports for access to the boot volume

```
lpardevicewwpn
LP17 a000 c05076e511803d60
LP17 a100 c05076e511803d9c
LP17 a200 c05076e511803dd8
LP17 a300 c05076e511803e14
```

You can use the devices in Example 22 for access to a set of shared cluster volumes.

Example 22 Devices

```
lpardevicewwpn
LP17 a001 c05076e511803d74
LP17 a101 c05076e511803db0
LP17 a201 c05076e511803dec
LP17 a301 c05076e511803e28
```

This approach requires additional zoning, host definitions, and LUN mappings for the devices in Example 22. However, it ensures that accidental access to host boot volumes is prevented by the explicit configuration of the host definitions on the storage controller.

This concludes the end-to-end configuration of the storage system:

- NPIV is enabled on both the host and storage ports.
- The host and storage NPIV ports are zoned to each other.
- The host is defined in the storage system and it has a LUN mapped to it.

Verifying SAN configuration by using HMC functions

Use the HMC web interface to examine the SAN configuration to verify that the configuration is correct.

Access the Channels view from the HMC by using the **Single Object Operations** function or, for z15 or later systems, by selecting the **Adapters** tab while the CPC is selected. Refer to the beginning of "Enabling NPIV on FCP channels" on page 10 for navigation instructions (Figure 53).

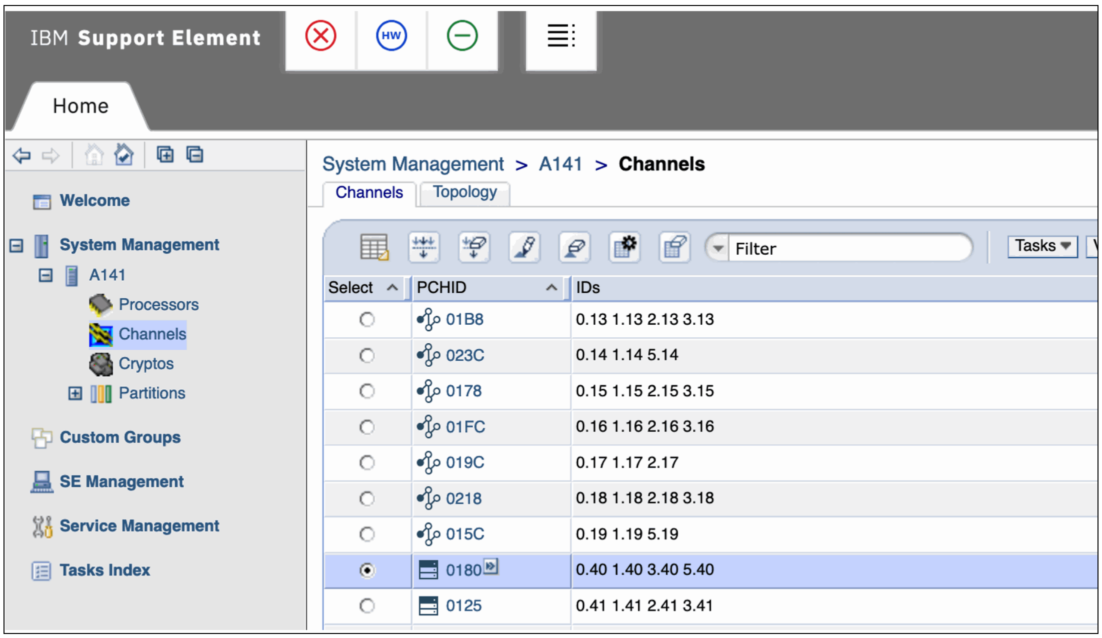


Figure 53 Channels view

With the channel selected, run the **Channel Problem Determination** action in the **Service** menu (Figure 54).

Figure 54 Channel Problem Determination action

From the displayed menu, select the LPAR for which you want to verify connectivity, and then click **OK** (Figure 55 on page 41).

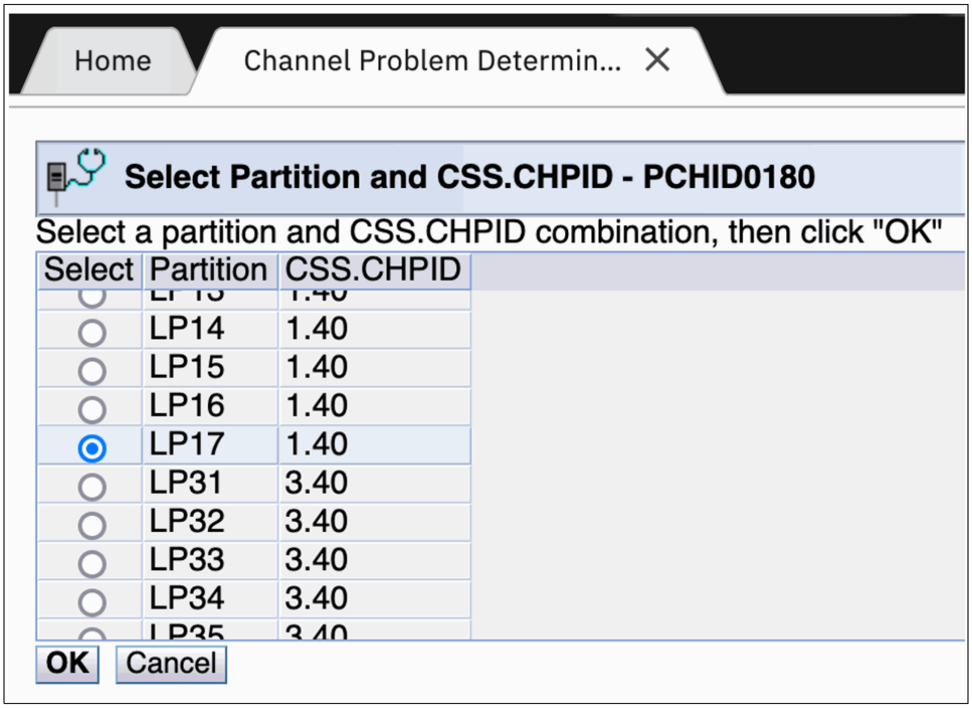


Figure 55 IBM Support Element menu

In the following menu, select **SAN explorer** and click **OK** (Figure 56).

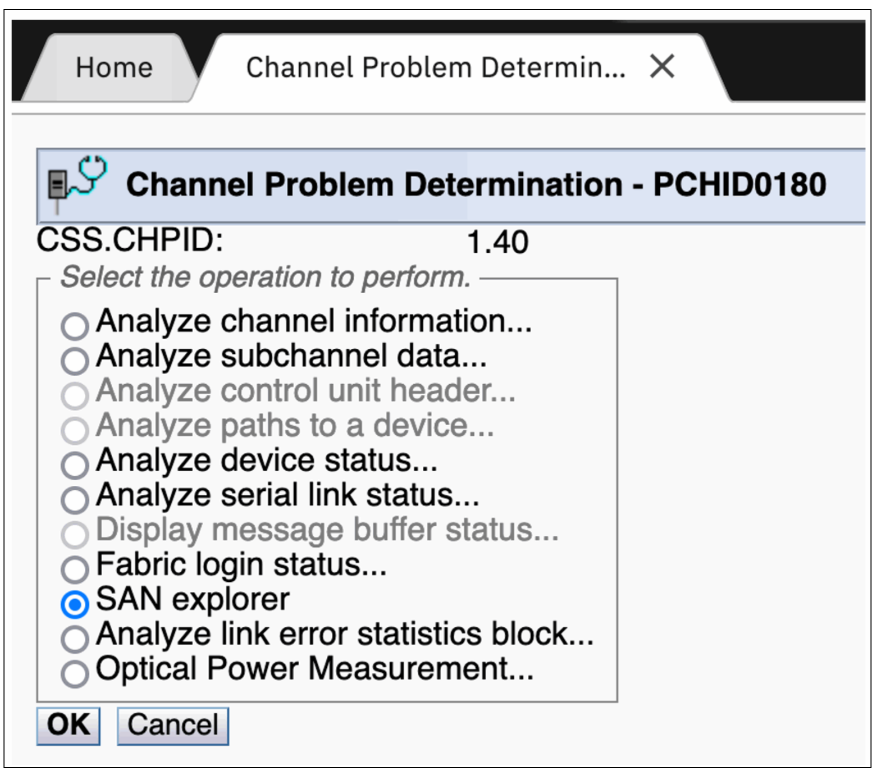


Figure 56 Select SAN explorer

At this point in this document, the examples diverge from a from-scratch configuration. These examples were generated using an existing, operational configuration rather than an unconfigured system. This difference is first apparent in Figure 57 on page 42.

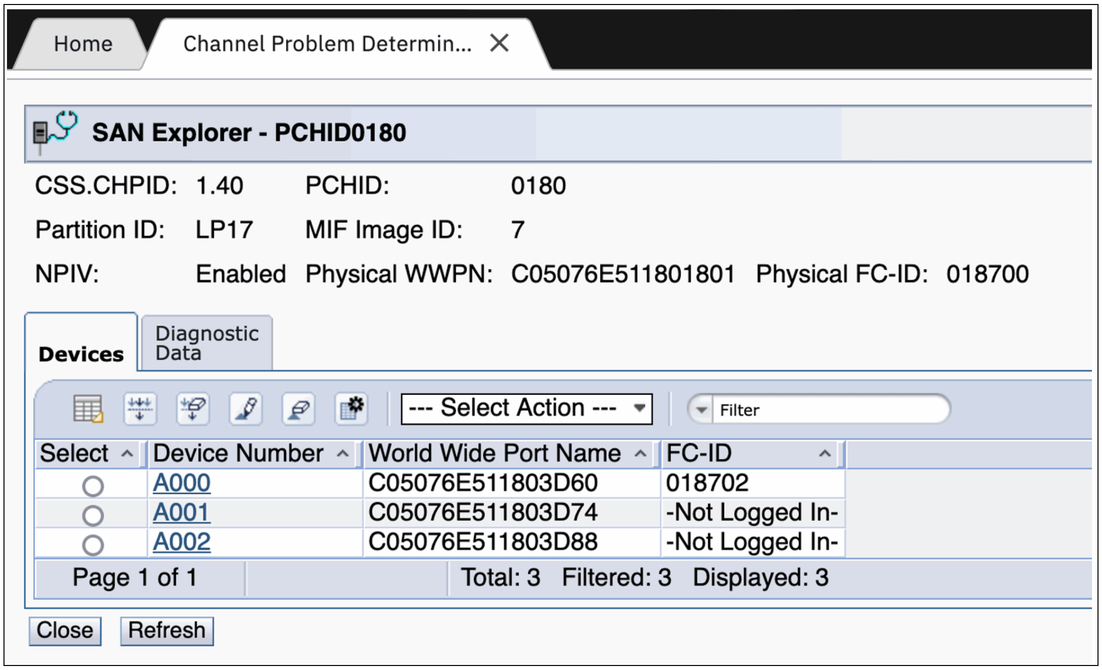


Figure 57 SAN Explorer menu

The presence of an FC-ID for device A000 indicates that it was actively registered with the SAN fabric at the time this panel was accessed. This implies that LP17 had a running operating system with device A000 varied online.

Before this point, the displayed panels used the current I/O configuration to populate menus and guide subsequent actions.

In Figure 57, selecting an LPAR's channel in SAN Explorer causes the system to examine the state of each FCP device on that channel within the LPAR. The NPIV WWPN of each device and its current registration status with the SAN fabric is then displayed. If the device is active and registered, its corresponding FC-ID is shown. Otherwise, the status `-Not Logged In-` is displayed.

If your LPAR is not running, you see a panel that looks similar to Figure 58 on page 43.

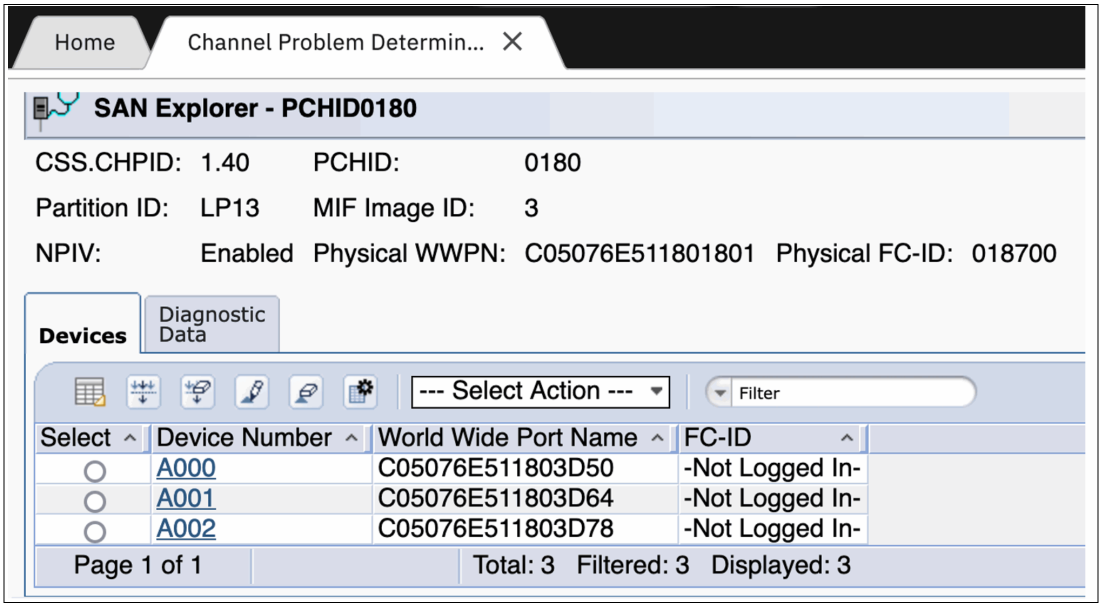


Figure 58   SAN Explorer menu

Going back to the LP17 panel with a running Linux OS in it, select the **A000** device number (Figure 59).

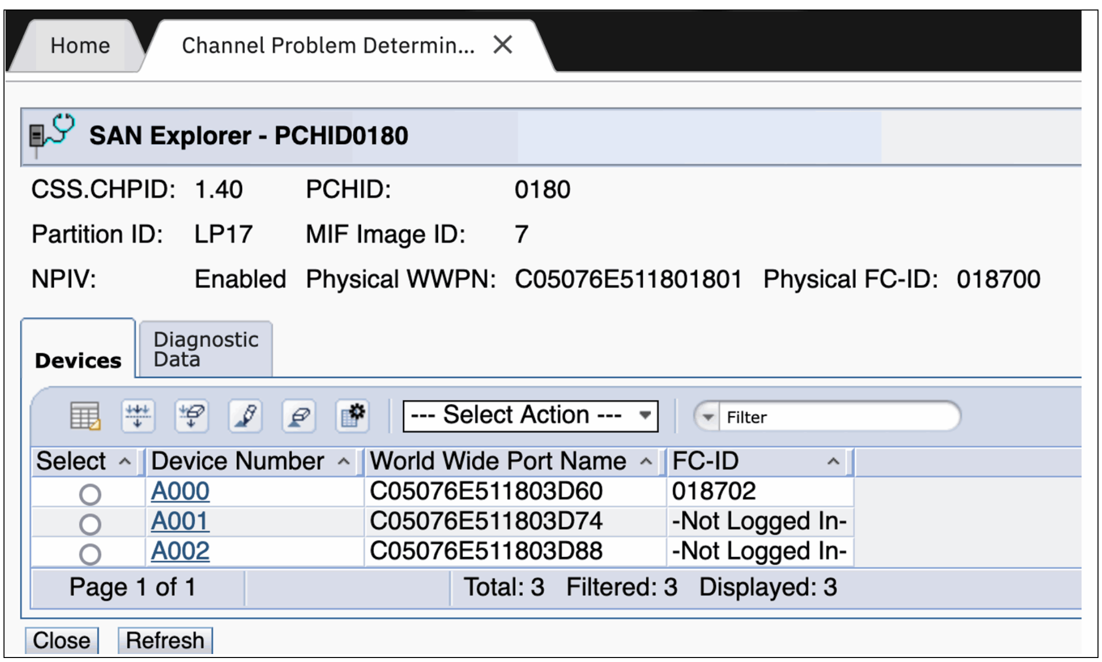
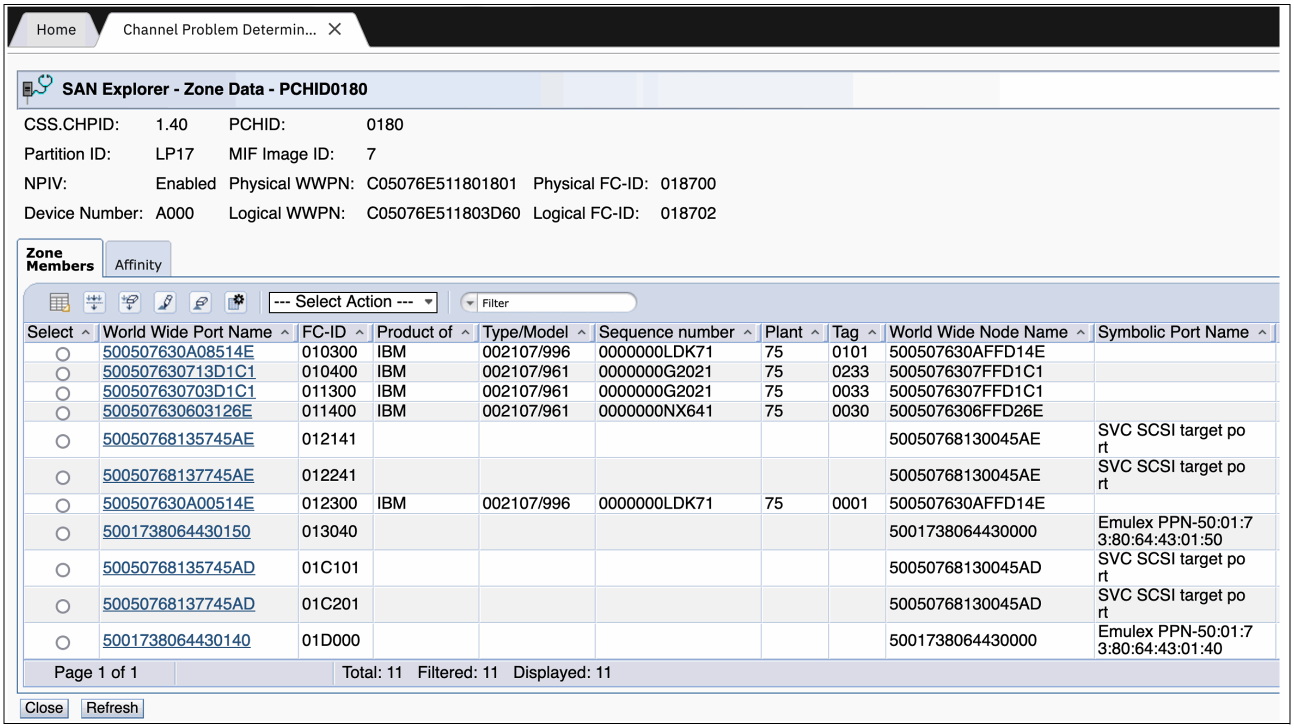


Figure 59   Click on the A000 device number

Selecting the LPAR's channel triggers the SAN Explorer function in the service element (SE) to initiate a fabric login from the LPAR's A000 device. The SE then queries the SAN name server for all accessible zoned ports before performing a fabric logout. A correctly zoned configuration results in a display similar to Figure 60 on page 44.



Home | Channel Problem Determin...

SAN Explorer - Zone Data - PCHID0180

CSS.CHPID: 1.40 PCHID: 0180

Partition ID: LP17 MIF Image ID: 7

NPIV: Enabled Physical WWPN: C05076E511801801 Physical FC-ID: 018700

Device Number: A000 Logical WWPN: C05076E511803D60 Logical FC-ID: 018702

Zone Members | Affinity

--- Select Action --- Filter

Select	World Wide Port Name	FC-ID	Product of	Type/Model	Sequence number	Plant	Tag	World Wide Node Name	Symbolic Port Name
○	500507630A08514E	010300	IBM	002107/996	0000000LDK71	75	0101	500507630AFFD14E	
○	500507630713D1C1	010400	IBM	002107/961	0000000G2021	75	0233	5005076307FFD1C1	
○	500507630703D1C1	011300	IBM	002107/961	0000000G2021	75	0033	5005076307FFD1C1	
○	500507630603126E	011400	IBM	002107/961	0000000NX641	75	0030	5005076306FFD26E	
○	50050768135745AE	012141						50050768130045AE	SVC SCSI target po rt
○	50050768137745AE	012241						50050768130045AE	SVC SCSI target po rt
○	500507630A00514E	012300	IBM	002107/996	0000000LDK71	75	0001	500507630AFFD14E	
○	5001738064430150	013040						5001738064430000	Emulex PPN-50:01:7 3:80:64:43:01:50
○	50050768135745AD	01C101						50050768130045AD	SVC SCSI target po rt
○	50050768137745AD	01C201						50050768130045AD	SVC SCSI target po rt
○	5001738064430140	01D000						5001738064430000	Emulex PPN-50:01:7 3:80:64:43:01:40

Page 1 of 1 Total: 11 Filtered: 11 Displayed: 11

Close Refresh

Figure 60 SAN Explorer - Zone data

The following example deviates from the ideal configuration that is presented in “An example configuration” on page 8. In contrast to the two zoned ports to the FS9500 illustrated in that example, the actual system configuration includes the following zone configuration:

- Four zoned ports to the FS9500
- Two ports to each of two DS8000 systems
- One port to a third DS8000
- Two ports to an IBM XIV®.

This discrepancy serves as an example of practical implementation that differs from ideal recommendations.

The recommended configuration for FS9500 connectivity is four paths to the first node and four paths to the second node. The zoning diagrams and procedures that are outlined in “Creating SAN zones to permit host ports to communicate with storage ports” on page 29" accurately depict this recommended configuration. The actual configuration shown Figure 60 provides an excessive number of paths, which is suboptimal for both the Linux multipath driver and the FS9500 itself.

If an NPIV WWPN is not zoned to any target ports, an empty panel, as shown Figure 61 on page 45, is displayed.

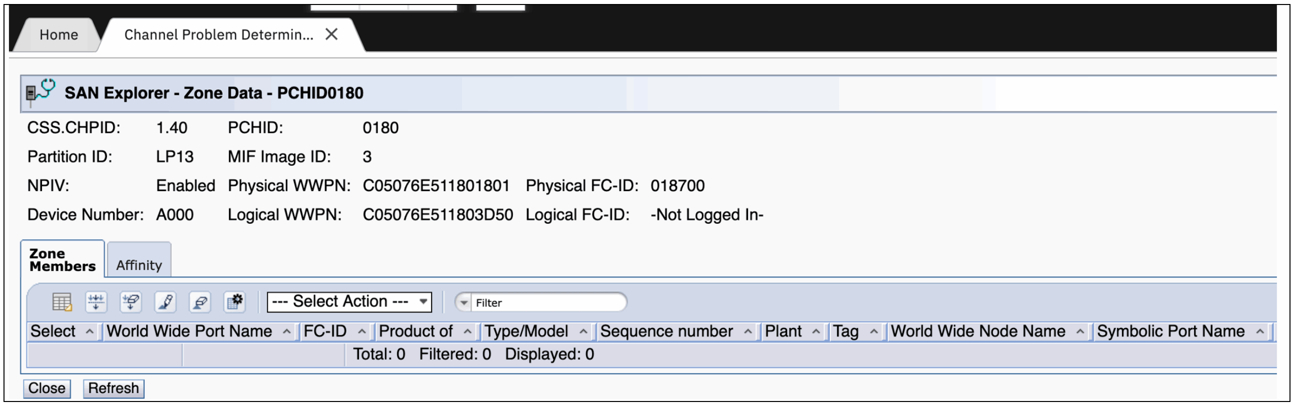
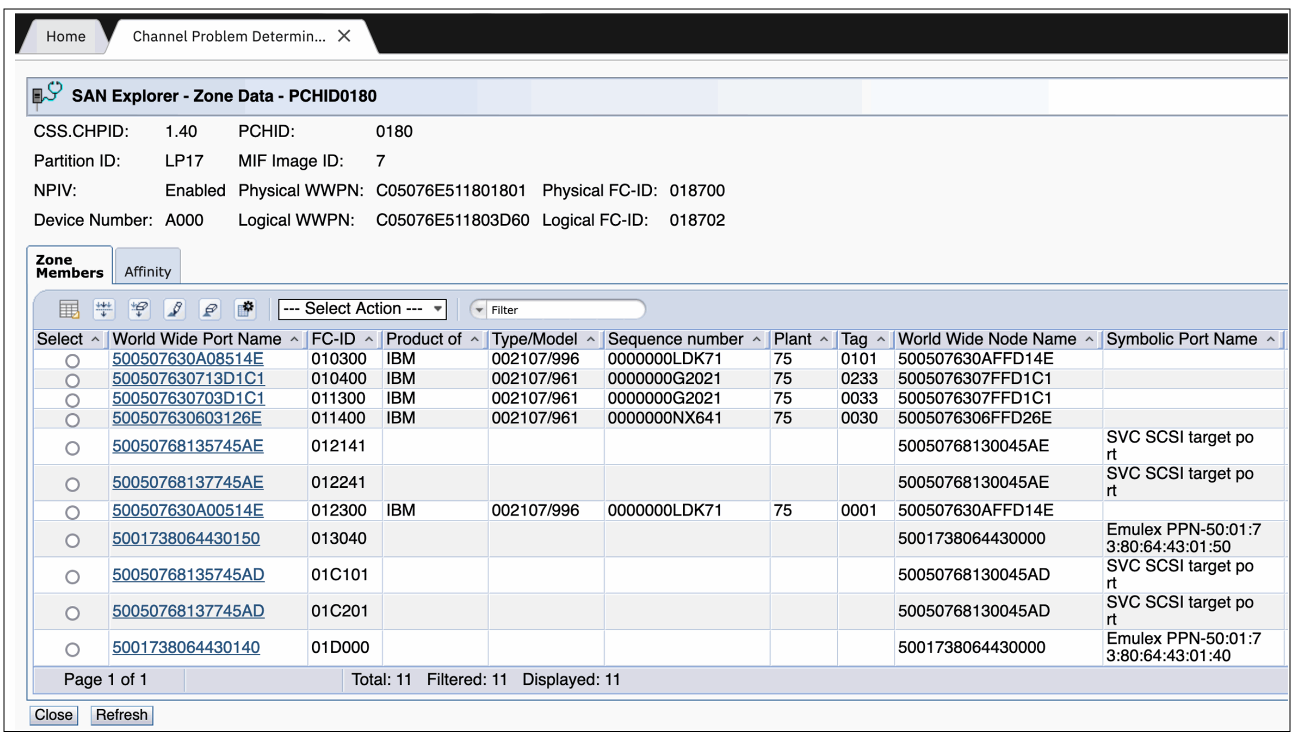


Figure 61 Empty panel

If the NPIV WWPN of the selected FCP device that is within the specified LPAR and channel is not zoned to any target ports, then no entries are listed in the panel. If the list in the panel includes unrecognized WWPNs, then the NPIV Logical WWPN for the selected device is incorrectly zoned and requires correction by the SAN administrator before you proceed.

Returning to the primary example, select one of the displayed target WWPNs corresponding to your storage controller to view the presented LUNs (Figure 62).



Select	World Wide Port Name	FC-ID	Product of	Type/Model	Sequence number	Plant	Tag	World Wide Node Name	Symbolic Port Name
○	500507630A08514E	010300	IBM	002107/996	0000000LDK71	75	0101	500507630AFFD14E	
○	500507630713D1C1	010400	IBM	002107/961	0000000G2021	75	0233	5005076307FFD1C1	
○	500507630703D1C1	011300	IBM	002107/961	0000000G2021	75	0033	5005076307FFD1C1	
○	500507630603126E	011400	IBM	002107/961	0000000NX641	75	0030	5005076306FFD26E	
○	50050768135745AE	012141						50050768130045AE	SVC SCSI target port
○	50050768137745AE	012241						50050768130045AE	SVC SCSI target port
○	500507630A00514E	012300	IBM	002107/996	0000000LDK71	75	0001	500507630AFFD14E	
○	5001738064430150	013040						5001738064430000	Emulex PPN-50:01:73:80:64:43:01:50
○	50050768135745AD	01C101						50050768130045AD	SVC SCSI target port
○	50050768137745AD	01C201						50050768130045AD	SVC SCSI target port
○	5001738064430140	01D000						5001738064430000	Emulex PPN-50:01:73:80:64:43:01:40

Figure 62 SAN Explorer - Zone Data

This action opens a list of the LUNs, if any, that the storage controller assigned to the host. The listed LUNs (Figure 63 on page 46) are determined by the host definition and the corresponding LUN map configured within the storage controller.

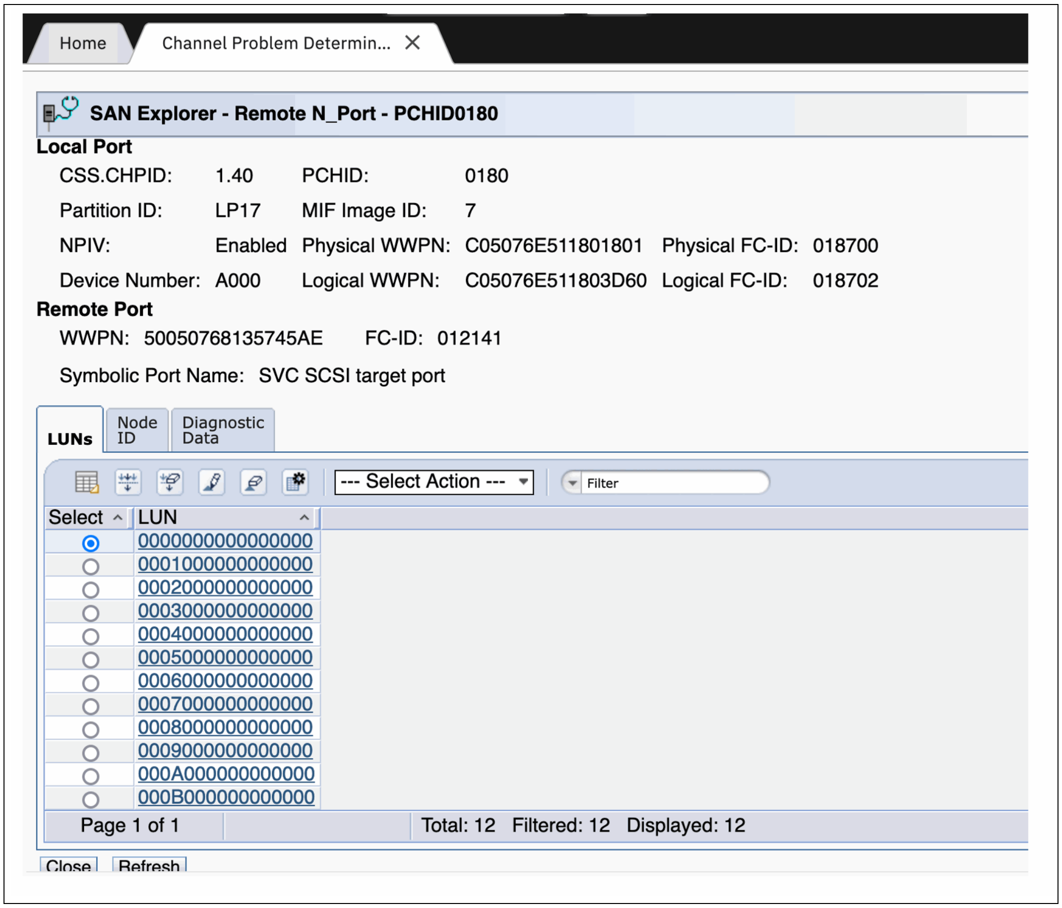


Figure 63 LUNs

You can see that the FlashSystem is presenting 12 LUNs: 0–B.

Note: IBM FlashSystem and SVC devices, by default, present LUNs sequentially, starting at LUN 0. On IBM LinuxONE and IBM Z, the LUN is represented by a 16-digit hexadecimal field, with the rightmost 12 digits set to 0. If needed, storage administrators can explicitly assign LUNs during volume mapping on the storage controller and can override the default sequential assignment.

In contrast, IBM DS8000 systems derive LUNs based on the Logical Control Unit (LCU) and unit address, padded with the hexadecimal value '40' and with only the rightmost 8 digits set to 0. For example, a volume at unit address 55 within LCU 33 on a DS8000 is assigned the LUN 4033405500000000. On DS8000 systems, the LUN presented to the host is determined by the system and cannot be modified by the storage administrator.

Select one of the 16 digit LUN numbers to open the next screen with details about that LUN (Figure 64 on page 47).

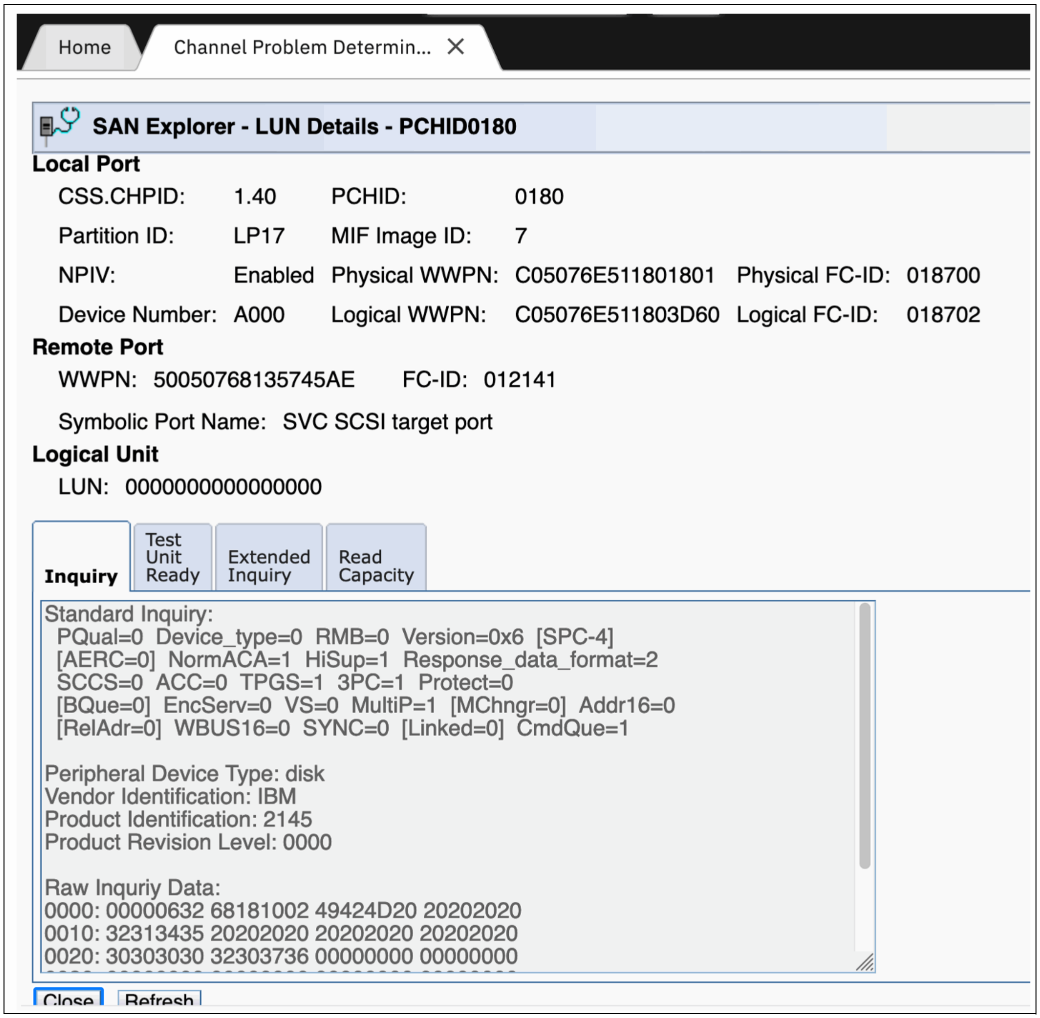


Figure 64 LUN details

In this screen, you can see the LUN details that SAN Explorer retrieved from the FlashSystem when you selected the LUN number on the previous panel. You can verify the Vendor and Product version strings in the Inquiry tab as shown in Figure 64.

If you click the **Test Unit Ready** tab, you see Ready in the big data field if the volume is ready for I/O.

If you click the Read Capacity tab, you see capacity details about the device (Figure 65 on page 48).

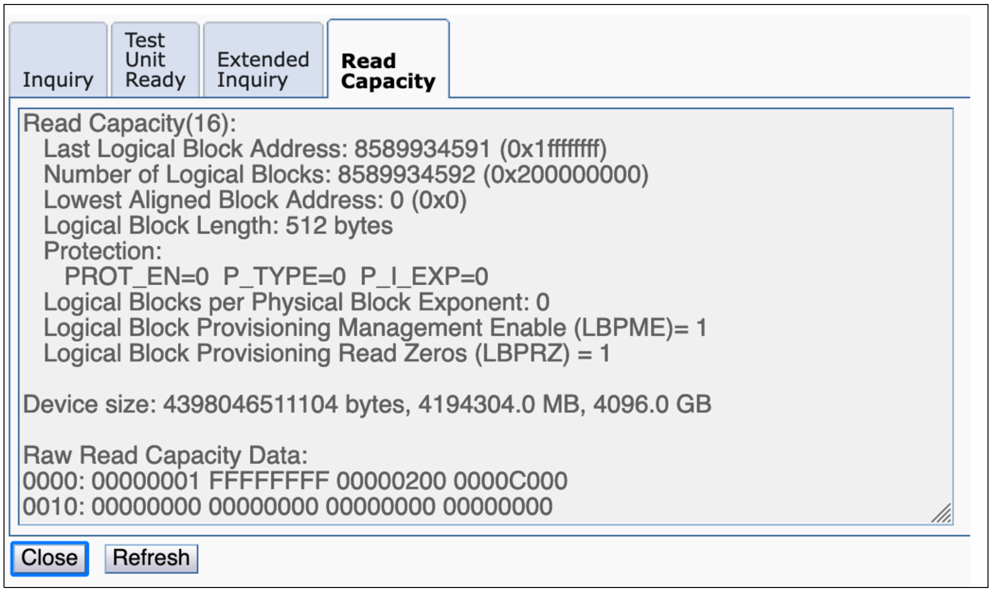


Figure 65 Read Capacity

Click **Close** until you see the Zone Data panel, and then click the other **WWPN** for your FlashSystem controller (Figure 66).

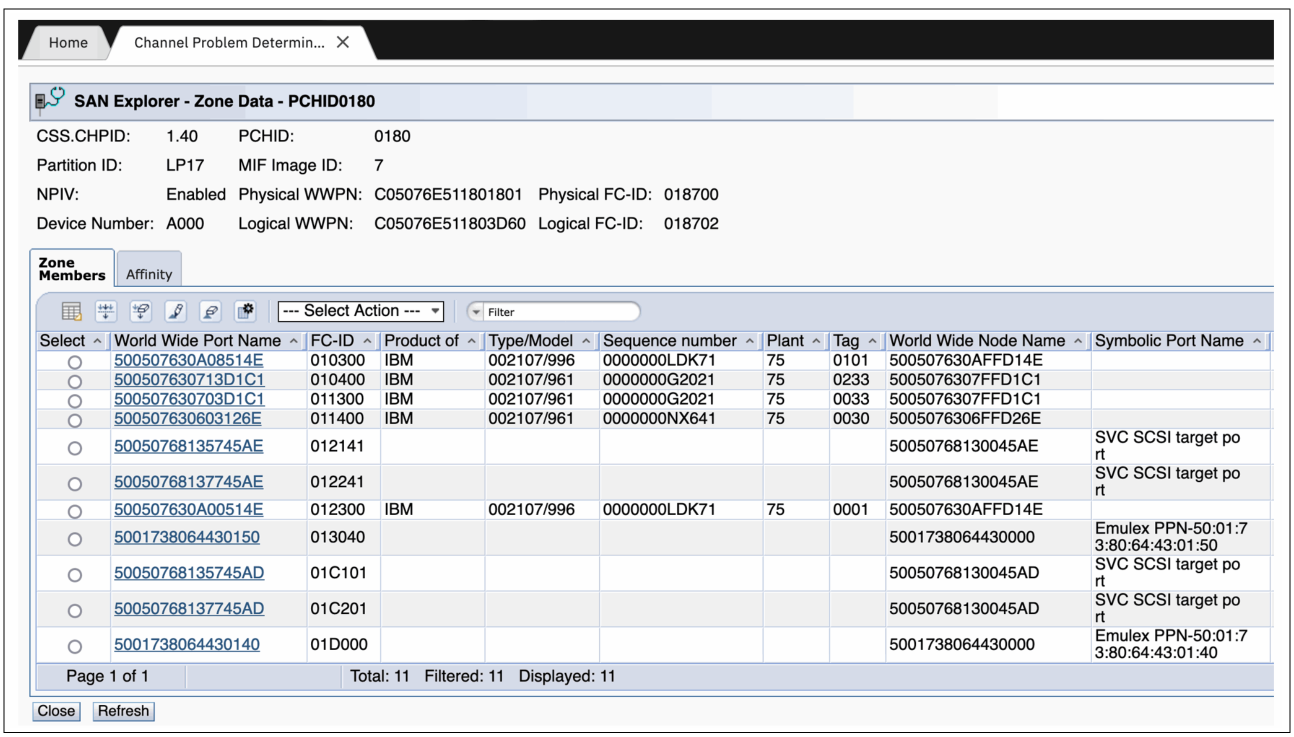


Figure 66 Zone Data panel

That opens the LUN list view (Figure 67 on page 49).

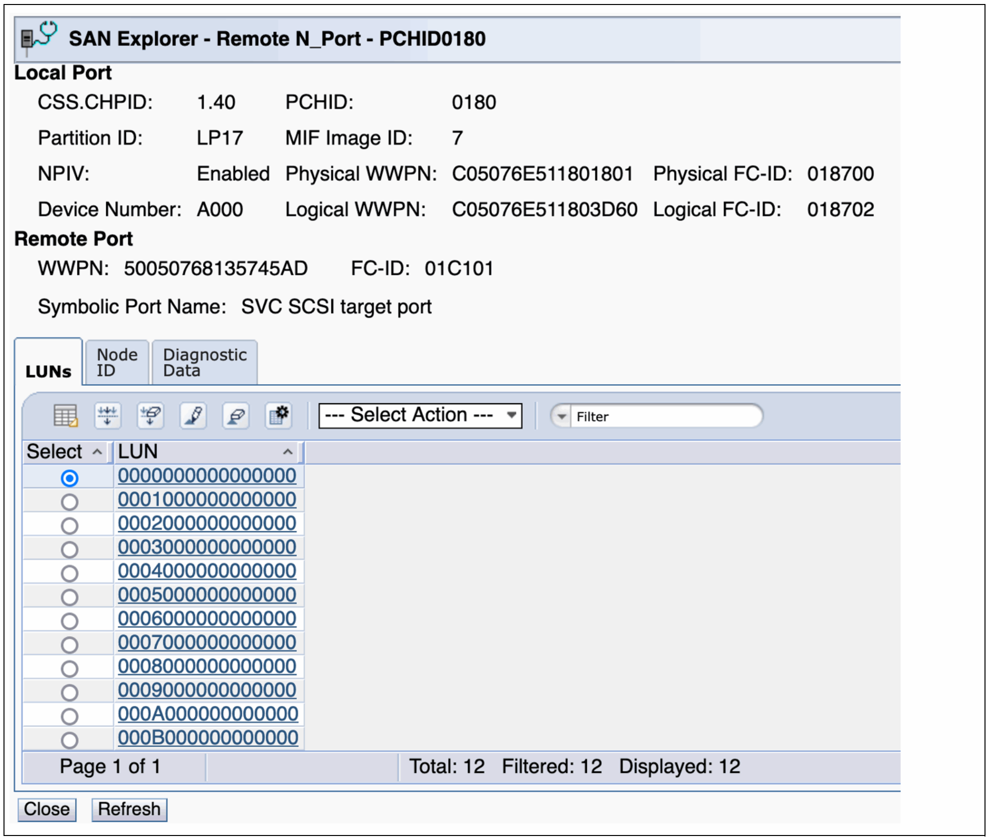


Figure 67 LUN list view

Verify that you see all your expected LUNs on this path to the other node as well.

Click **Close** or **Cancel** until you see the Channel or the Adapters listing, select another FCP Channel, and repeat the process to make sure that its ports are all zoned as required. For each storage WWPN, also compare the LUNs provided by the storage controller to make sure they are consistent across all configured paths.

At this point, you used the HMC San Explorer task to verify all the configured zones in the SAN switches and the host definitions and LUN maps in the storage controller.

Summary

This Redpaper publication included the following topics:

- An overview of FCP, and some of how it contrasts versus IBM FICON
- An example I/O configuration of FCP channels shared by multiple LPARs
- An example of how to enable NPIV on FCP channels, and discover the generated WWPNs
- An example of how to discover the correct WWPNs to use for zoning on an FS9500
- An example for both Cisco and Brocade zoning by using the new zoning features
- An example for creating Host definitions and LUN maps on a FS9500
- An example of using the SAN Explorer HMC function to ensure that all of the previous items were done as expected

At this point, the LPARs SAN configurations are usable for Linux. For more information, refer to your Linux documentation.

Appendix - Handy switch command cheat sheets

You can use the following often-used commands for Brocade and Cisco switches.

Brocade command cheat sheet:

Check if FCP dev is logged in (This also shows if the FCP device has an alias):

```
nodefind <fcp_dev_wwpn> or <alias>
```

Get WWPN for a given alias:

```
alishow <alias> <---Can accept wildcards, for example, alishow *fs95fp*
```

Find zone config:

```
zoneshow <zone_name> <---Can accept wildcardsfor example zoneshow *fs95fp*
```

Find if fcp_dev is in a zone:

```
zoneshow | grep <wwpn>
```

Find if alias is in a zone:

```
zoneshow --alias <alias_name>
```

Alias management:

- Create alias:

  ```
  alicreate "<alias-name>", "<fcp_dev_wwpn>"
  ```

- Change alias:

  ```
  zoneobjectrename "<old_alias_name>", "<new_alias_name>"
  ```

- Remove alias:

  ```
  alidelete "<alias-name>"
  ```

Create zone:

```
zonecreate --peerzone <zone_name> -principal
"<storage_controller_1;storage_controller_2> -members "<alias_name_1;alias_name2>"
```

Add zone to zoneset:

```
cfgadd <zoneset>,<newly_added_zone_name>
```

Save changes:

```
cfgsave -f
cfgenable <zoneset>
```

Check if zone is in active config:

```
cfgactvshow | grep <zone_name>
```

Cisco command cheat sheet:

Check if FCP dev is logged in:

```
show fcs port pwwn <wwpn> vsan <vsan_number>
```

Check if alias exists for a FCP dev:

```
show device-alias database | grep <your:wwpn:number>
```

Get WWPN for an alias:

```
show device-alias name <alias_name>
```

Find if alias or fcp_dev is in a zone:

```
show zone | grep <alias_name or wwpn>
```

Get configuration for a zone:

```
show zone name <zone_name>
```

Alias management:

```
config -t
device-alias database
```

- Create alias:

  ```
  device-alias name <alias-name> <your:wwpn:number>
  ```

- Change alias:

  ```
  device-alias rename <old-alias> <new-alias>
  ```

- Remove alias:

  ```
  no device-alias name <alias-name>
  ```

- Save alias changes:

  ```
  device-alias commit
  ```

Create zone:

```
config -t
zone name <zone-name> vsan <vsan_number>
```

Add members to a zone:

```
member device-alias <alias-name> init <--(if the alias is an FCP device) target
(if the alias is a npiv storage device)
```

Remove members to zone:

```
no member device-alias <alias-name>
```

Adding zones to a zoneset and activating changes:

```
zoneset <zoneset> vsan <vsan_number>
member <new_zone_name>
zoneset activate name <zoneset> vsan <vsan_number>
```

Author

This paper was produced by a team of specialists from around the world.

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